



Create a CI/CD Pipeline on Azure

Week 10 Practical

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Preamble

Azure DevOps by Microsoft Azure is one of the leading tools that automate CI/CD's process and, in turn, supports automatic builds and code projects to make them available to others. The Azure pipelines combine Continuous Integration (CI) and Continuous Delivery (CD) to consistently test and build the code and ship it to the target environment.

Azure Pipelines

The Azure CI/CD pipeline simplifies continuous integration and continuous delivery (CI/CD) in the application development process. You can start from the source stage with existing code on GitHub or on-premise containers. The Azure Repos can maintain a central repository, and the Azure pipelines maintain build and release pipelines for the given project. The Azure DevOps CI/CD process is a crucial process with all the required dev services.

Apart from continuous integration and continuous deployment with Azure DevOps, these pipelines are used to construct build-deploy-test workflows used mainly in continuous testing (CT). This tests the changes in a fast and scalable routine.

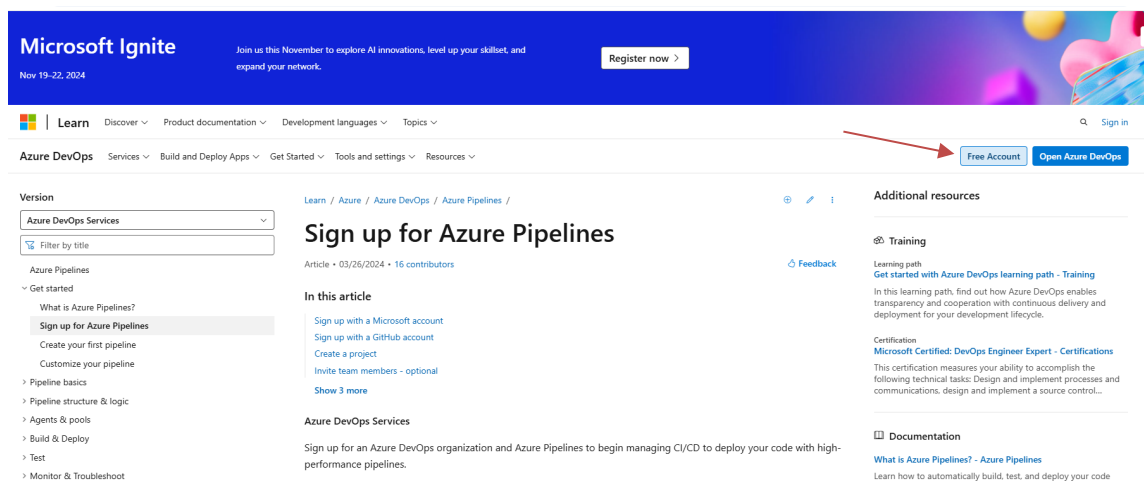
Creating an ASP .NET CI/CD Project using an Azure Pipeline

Azure DevOps project simplifies the setup of CI/CD pipelines in Azure. You can go with the existing code or create a sample application as a new Azure DevOps project.

Step-1: Prerequisites

Make sure you have the following items:

1. A GitHub account where you can create a repository. [Create one for free.](#)
2. An ability to run pipelines on Microsoft-hosted agents. If you haven't done this before then please fill out the following form <https://aka.ms/azpipelines-parallelism-request>
It will take 2-3 days to get the response and approve your grant.
3. An Azure DevOps organization using this link: [Create one for free.](#)



Click on 'Free Account'

On next page, click on 'Start Free' under Azure pipelines:

azure.microsoft.com/en-us/pricing/details/devops/azure-devops-services/

Microsoft | Azure | Solutions | Pricing | Partners | Resources | Learn | Support | Contact Sales | [Get started with Azure](#) | [Sign in](#)

Pricing for Azure DevOps

[Azure DevOps Services](#)
[Azure DevOps Server](#)

Prices are estimates only and are not intended as actual price quotes. Actual pricing may vary depending on the type of agreement entered into with Microsoft, date of purchase, and the currency exchange rate. Prices are calculated based on US US dollars and converting. Consider pricing per user that are reflected in the first business day prior to the last business day of the previous month end. The first business day prior to the end of the month that is a business day in that country. The rate setting day is generally the day immediately preceding the first business day. This rate applies to all transactions during the upcoming month. Sign in to the [Azure account](#) to see pricing based on your current usage with Microsoft. Contact an [Azure sales rep](#) for more information on pricing or to request a price quote. See [Azure pricing](#) for more information.

Currency: [United States - Dollar \(\\$ USD\)](#)

INDIVIDUAL SERVICES

Azure Pipelines

1 Free Microsoft-hosted CI/CD
1 Free Self-hosted CI/CD

[Start free](#)

- 1 Microsoft-hosted job with 1,800 minutes per month for CI/CD and 1 self-hosted job with unlimited minutes per month
- \$40 per extra Microsoft-hosted CI/CD pipeline job and \$15 per extra self-hosted CI/CD pipeline job with unlimited minutes

Microsoft-hosted

Azure Artifacts

2 GB free, then starting at \$2 per GB

[Start free](#)

- Industry-leading NuGet Server
- Support for Maven, npm, and Python packages
- Uploads sources to help protect open-source dependencies
- Integrated with Azure Pipelines
- Scalable access controls

First 2 GB free

GitHub Advanced Security for Azure DevOps

\$49 per contributor per month

[Activate now](#)

- Full suite of security tools native to Azure DevOps
- Secret scanning
- Code scanning
- Dependency scanning

Total contributors

USER LICENSES

Basic Plan

First 5 users free, then \$5 per user per month

[Start free](#)

- Azure Pipelines includes the free offer from INDIVIDUAL SERVICES
- Azure Boards: Work item tracking and Kanban boards
- Azure Repos: Unlimited private Git repos
- Azure Artifacts: 2 GB free per organization

First 5 users free. For additional users, see above.

Basic - Test Plans

\$12 per user per month

[30-day Free Trial](#)

- Includes all Basic plan features
- Test planning, tracking & execution
- Browser-based tests with annotation
- Rich-client test execution
- User acceptance testing
- Centralized reporting

\$12 per user per month

Select your Ulster University account



Pick an account



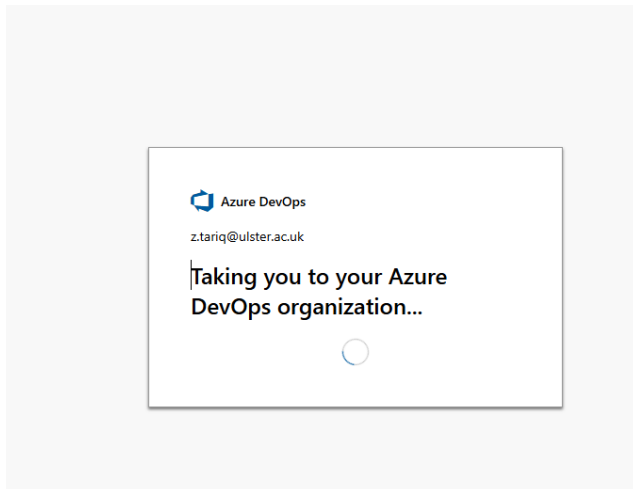
Tariq, Zeeshan
z.tariq@ulster.ac.uk
Signed in

⋮

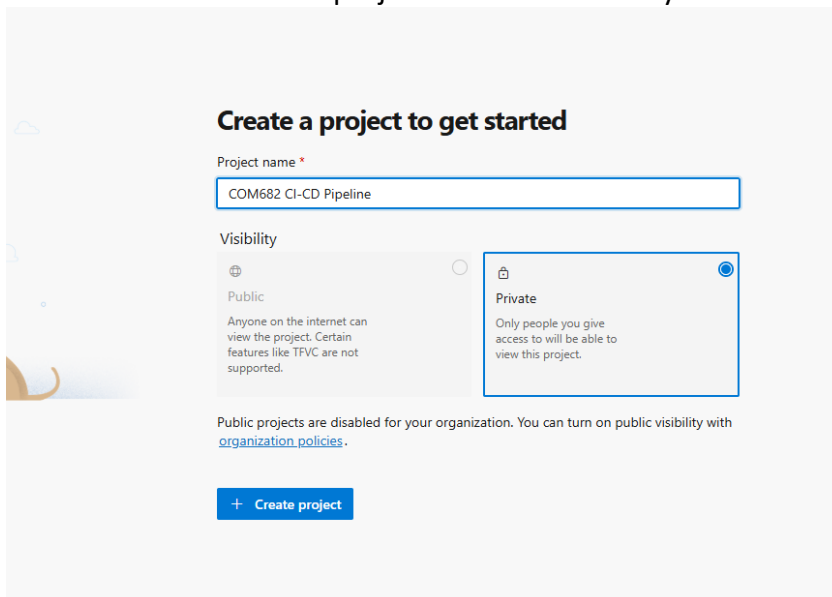


Use another account

Select the name, select the project hosting (default is preferred) and after verification of press 'continue'



You will be asked for the project name and visibility

A screenshot of the "Create a project to get started" form in Azure DevOps. The form has a title "Create a project to get started". Below the title, there is a "Project name" field with a red asterisk, containing the text "COM682 CI-CD Pipeline". Below the project name field, there is a "Visibility" section with two radio buttons. The "Public" option is selected, and the "Private" option is also visible. The "Public" option description says "Anyone on the internet can view the project. Certain features like TFVC are not supported." The "Private" option description says "Only people you give access to will be able to view this project." Below the visibility options, there is a note: "Public projects are disabled for your organization. You can turn on public visibility with [organization policies](#)." At the bottom of the form, there is a blue button with a plus sign and the text "Create project".

Click on Create Project

Once this step is completed you can view your organization and project at <https://aex.dev.azure.com/me?mkt=en-US>

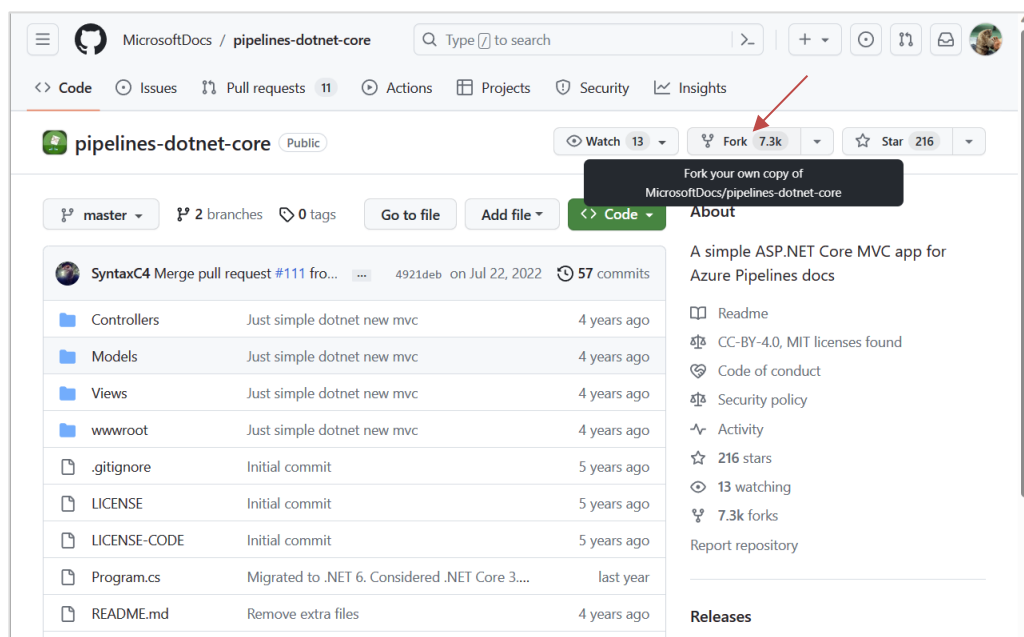
Step-2: Cloning a target repository into your GitHub Account

In order to establish a CI/CD pipeline we first need to have a source code repository which can be compiled and deployed by a pipeline. As writing the source code for such a repository is outside of the remit of this practical exercise, we will be cloning an existing one.

Specifically, we will fork the repository at :

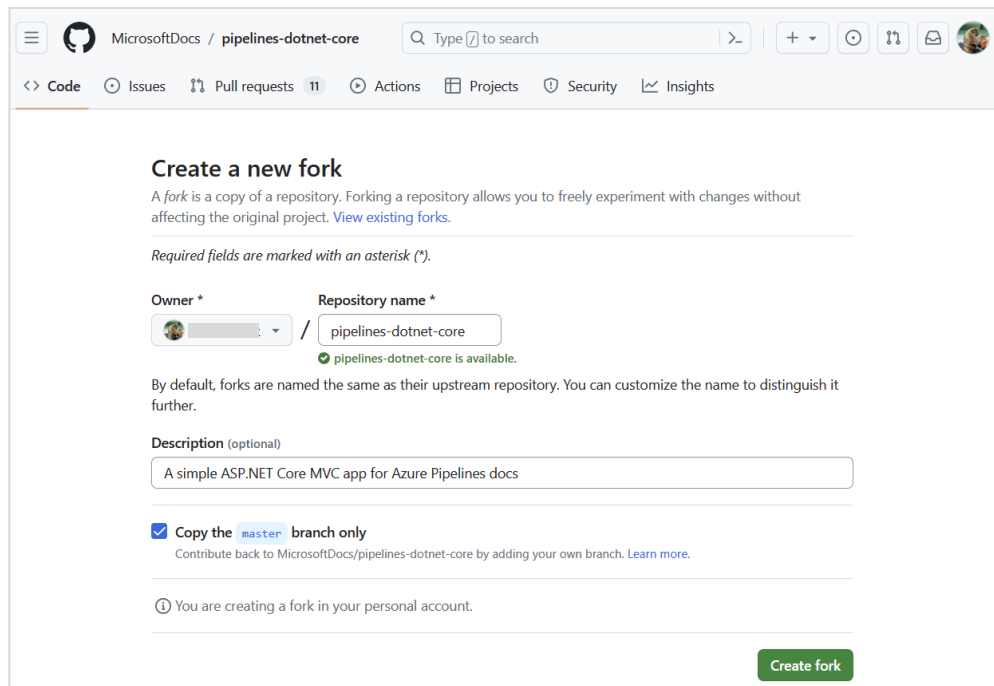
<https://github.com/MicrosoftDocs/pipelines-dotnet-core>

Ensure you are signed into your GitHub account and then browse to the above URI.



Click the fork button to initiate the forking process. This will present the start of the fork wizard which will allow you:

1. To select the owner - which should be your account
2. Provide a repository name - we will operate with the default for this as we have no conflict
3. Provide a description.
4. Specify if you want to copy the master branch only



MicrosoftDocs / pipelines-dotnet-core

<> Code Issues Pull requests 11 Actions Projects Security Insights

Create a new fork

A fork is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project. [View existing forks.](#)

Required fields are marked with an asterisk (*).

Owner * / Repository name * pipelines-dotnet-core
 pipelines-dotnet-core is available.

By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.

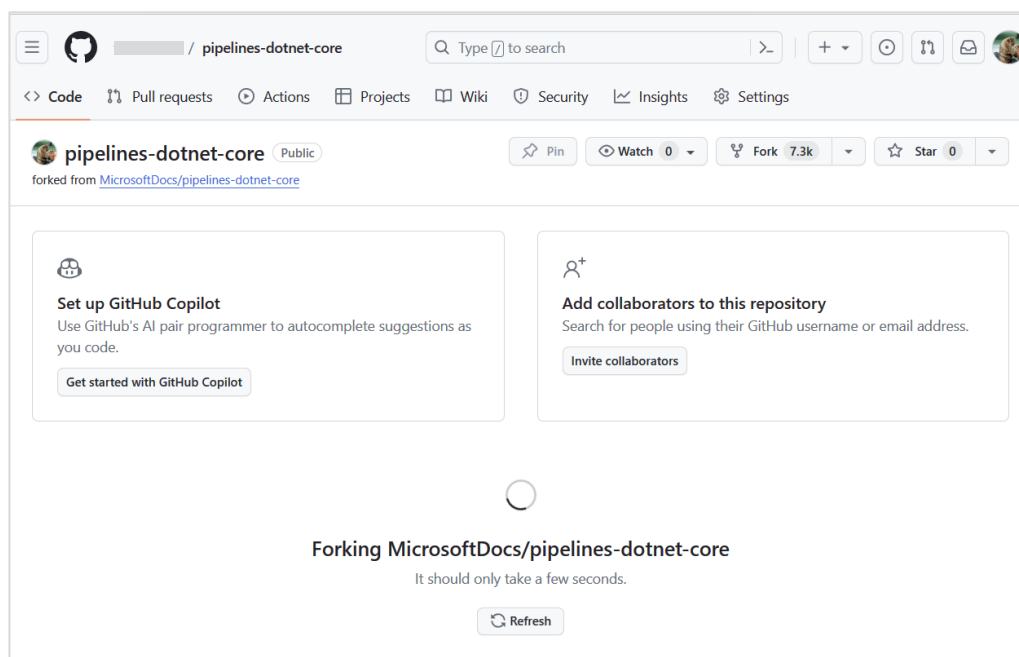
Description (optional)
 A simple ASP.NET Core MVC app for Azure Pipelines docs

☒ Copy the master branch only
 Contribute back to MicrosoftDocs/pipelines-dotnet-core by adding your own branch. [Learn more.](#)

You are creating a fork in your personal account.

Create fork

Specify these options as desired then click “create fork”.



pipelines-dotnet-core (Public) Pin Watch 0 Fork 7.3k Star 0

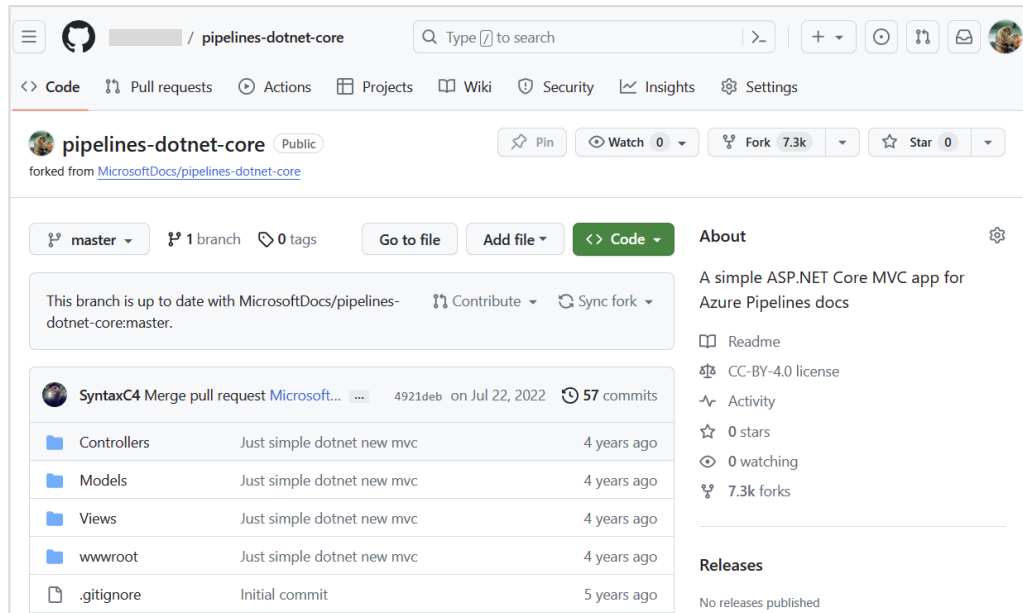
forked from [MicrosoftDocs/pipelines-dotnet-core](#)

Set up GitHub Copilot
 Use GitHub's AI pair programmer to autocomplete suggestions as you code.
[Get started with GitHub Copilot](#)

Add collaborators to this repository
 Search for people using their GitHub username or email address.
[Invite collaborators](#)

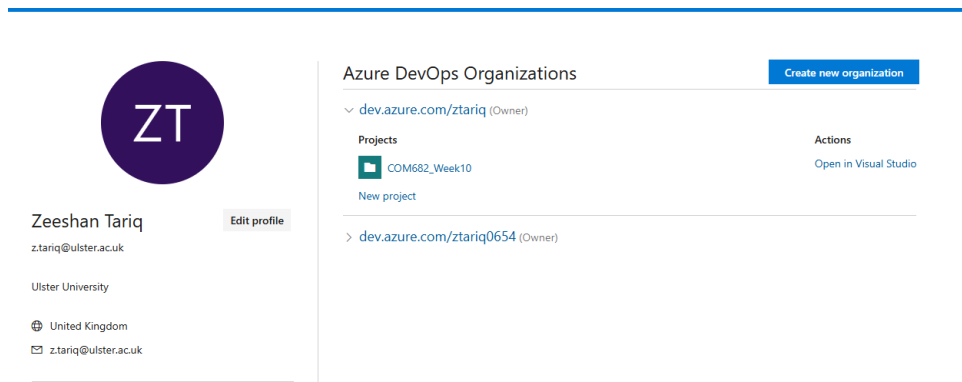
Forking MicrosoftDocs/pipelines-dotnet-core
 It should only take a few seconds.
[Refresh](#)

After a short process, the repository will be cloned into your GitHub account.

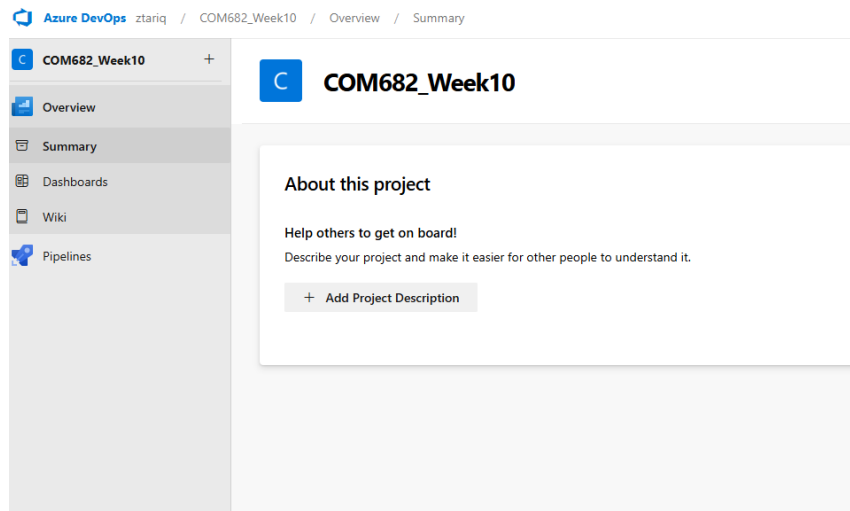


Step-3: Setting up a DevOps project on Azure

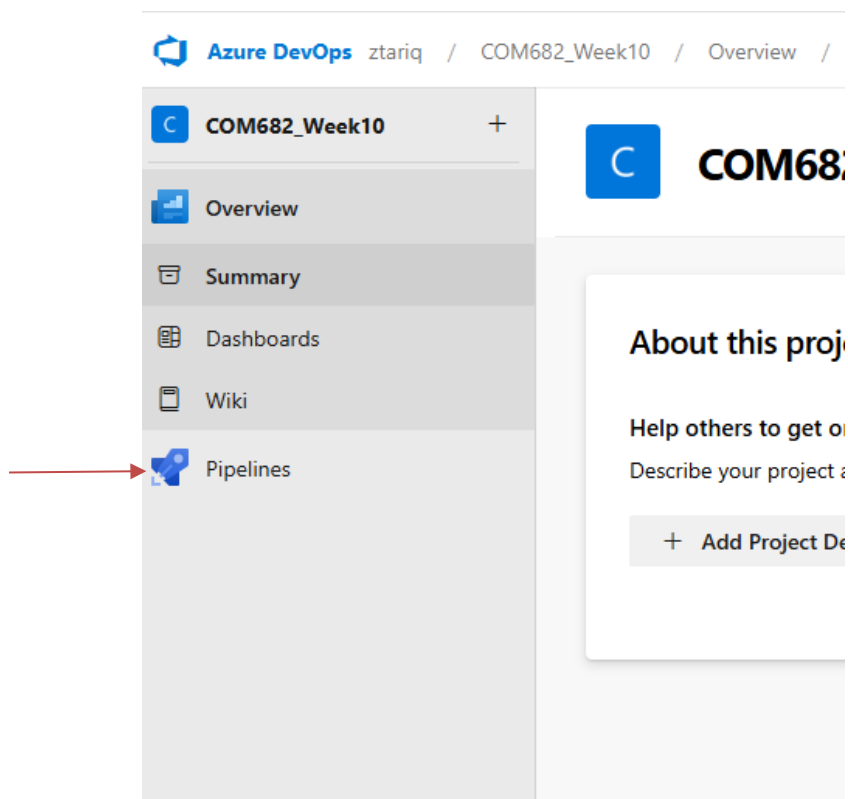
Access your Azure DevOps Organisation you created previously via <https://aex.dev.azure.com/me?mkt=en-US>. In this case, we have set up the ztariq organisation and project name is COM682_week10 for this practical.



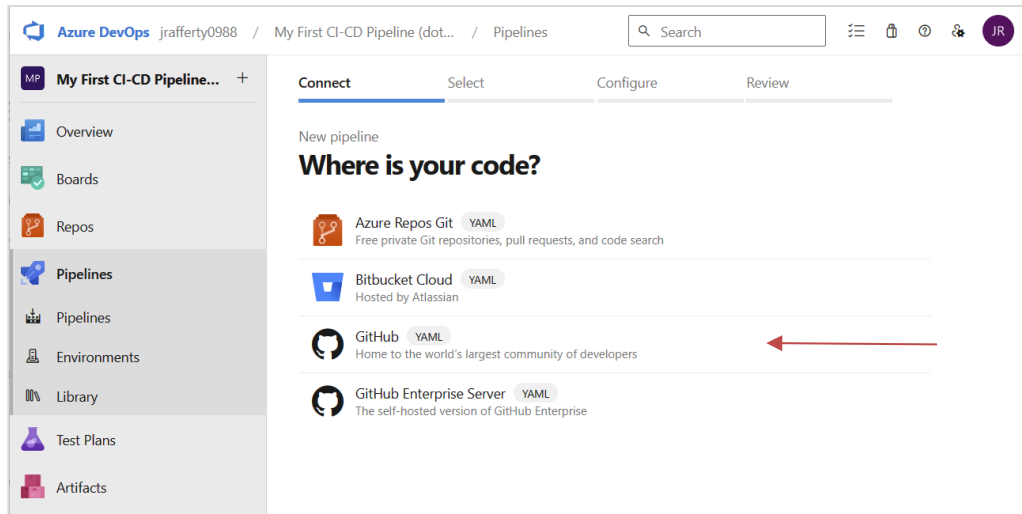
Click on your project



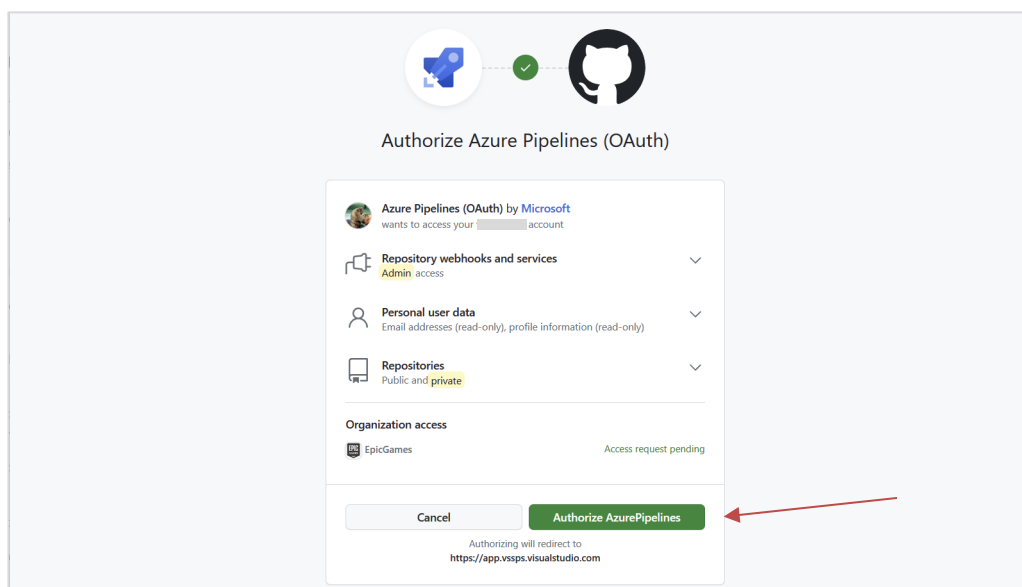
Next, select the pipelines icon  you will be presented with the pipelines splash page, select “Create Pipeline”.



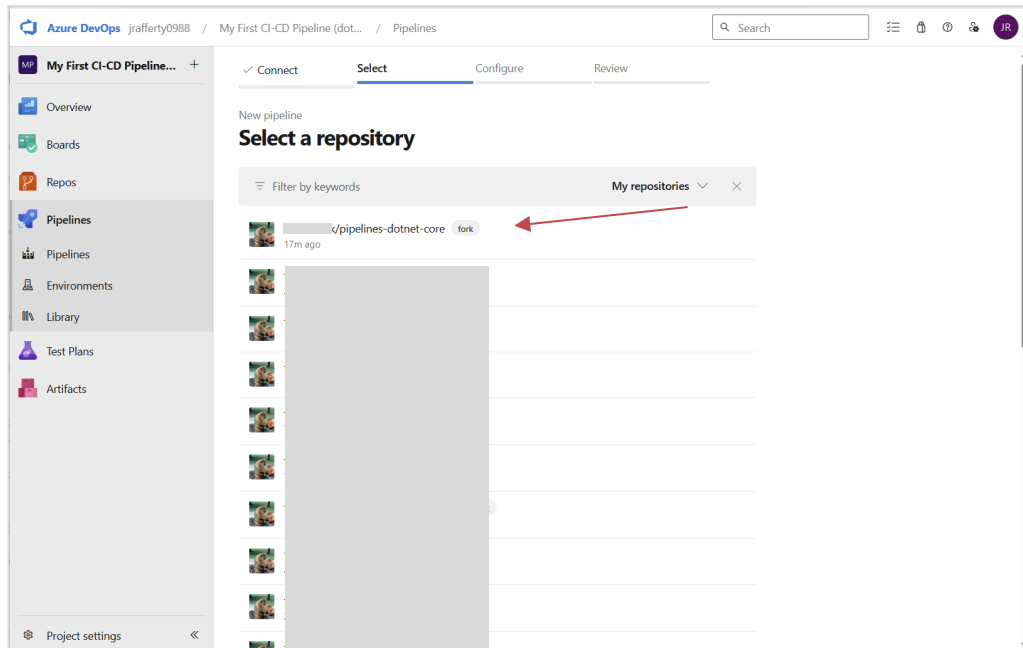
The pipeline integrator will first ask you where your code is located, we have chosen GitHub so will select that option. Note the other integrations available.



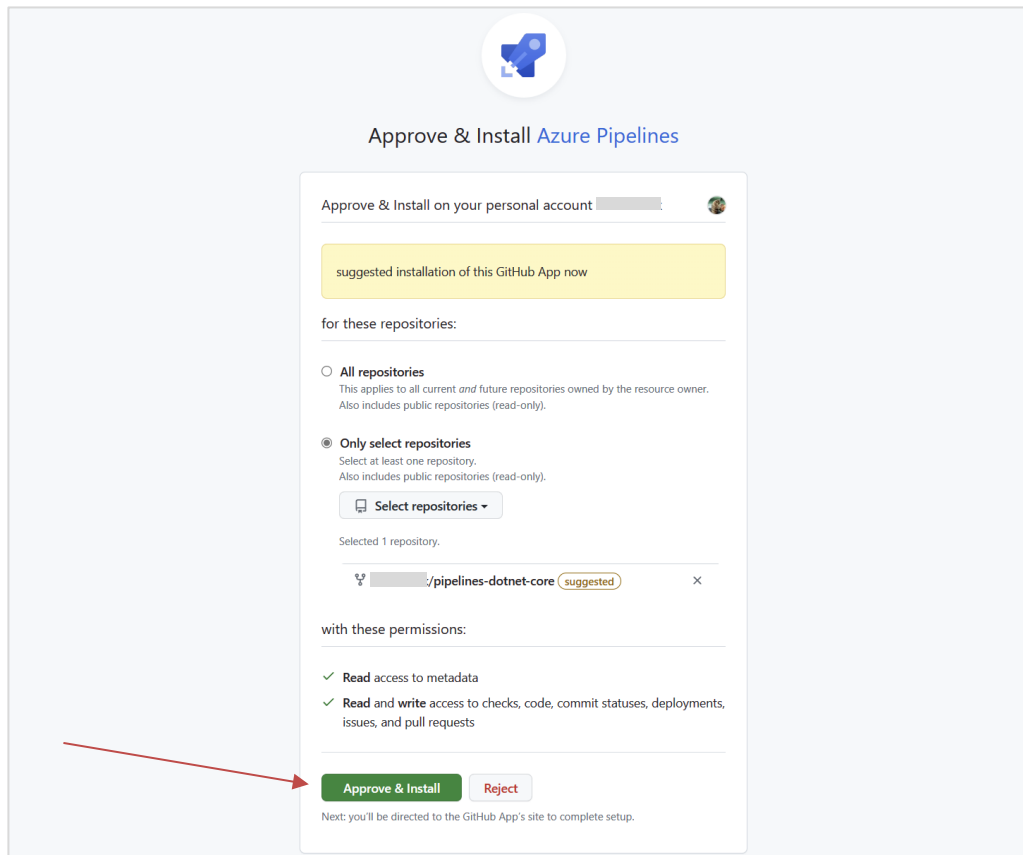
You will be asked to authorise access to your GitHub account from Azure Pipelines , you need to provide this permission to continue.



You will then be presented with a list of the repositories associated with your GitHub account. Please select the one you created earlier in this tutorial, in this case it is pipelines-dotnet-core

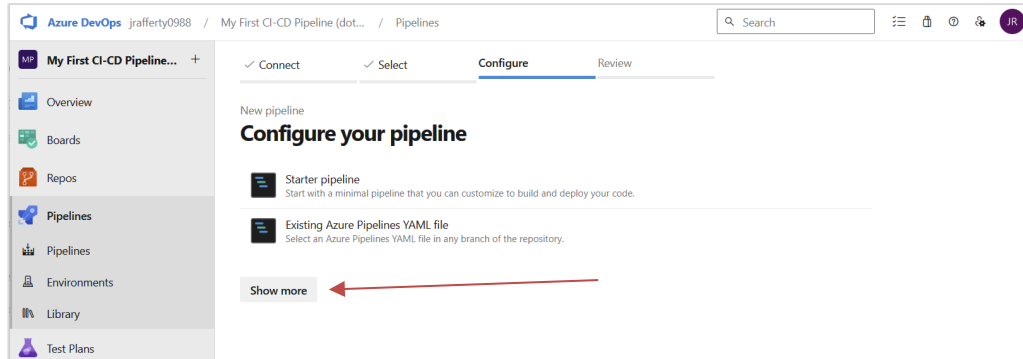


You will then be asked to approve and install Azure Pipelines into that repository, you need to click approve and install to continue. Once you select this option you may need to re-authenticate to GitHub and/or Azure.

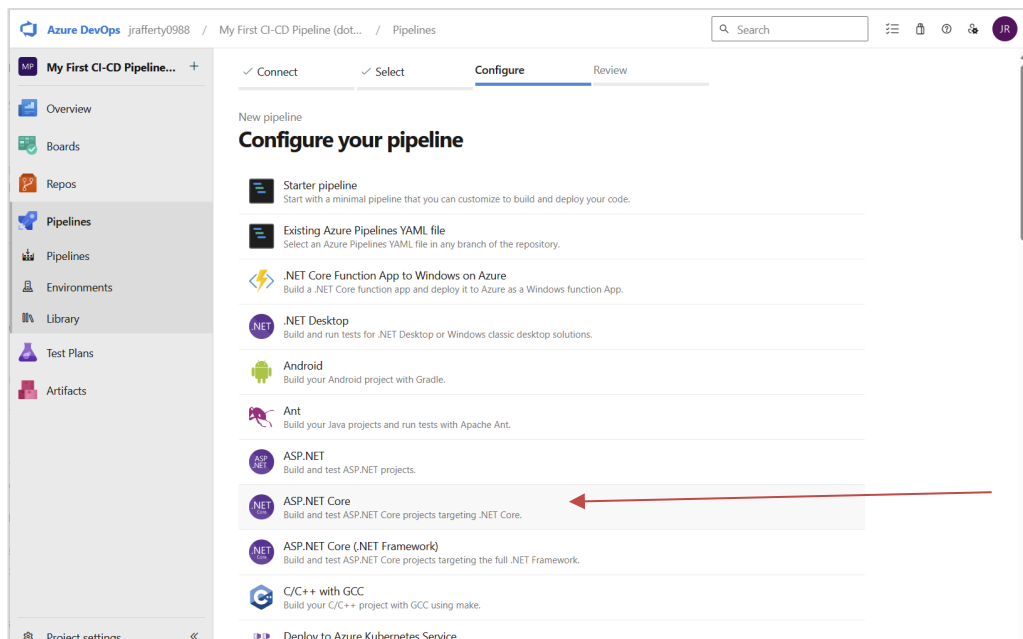


When the process is complete, you will be prompted to configure a pipeline, for this activity we should

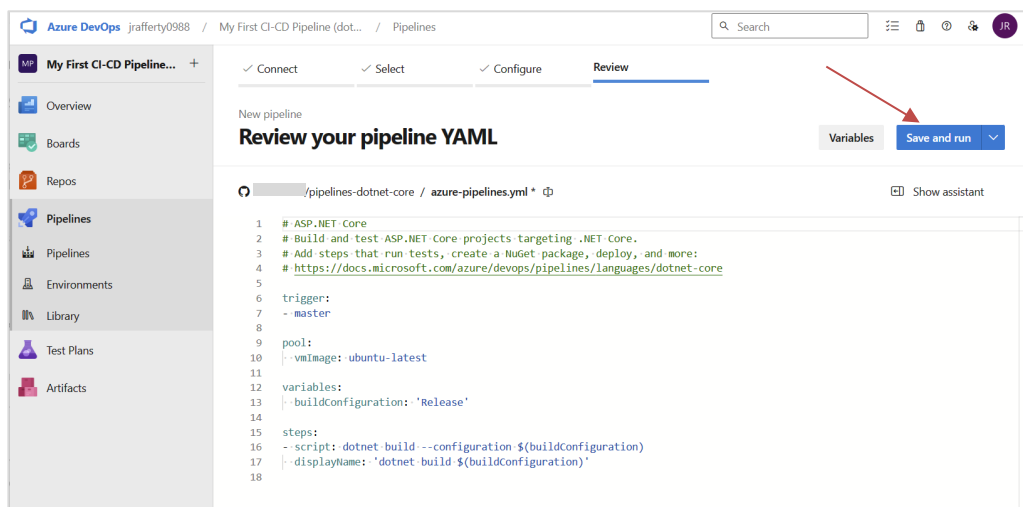
select show more as the template we wish to you is not present.



As this project is based upon ASP.NET core we should then select the ASP.NET Core template.

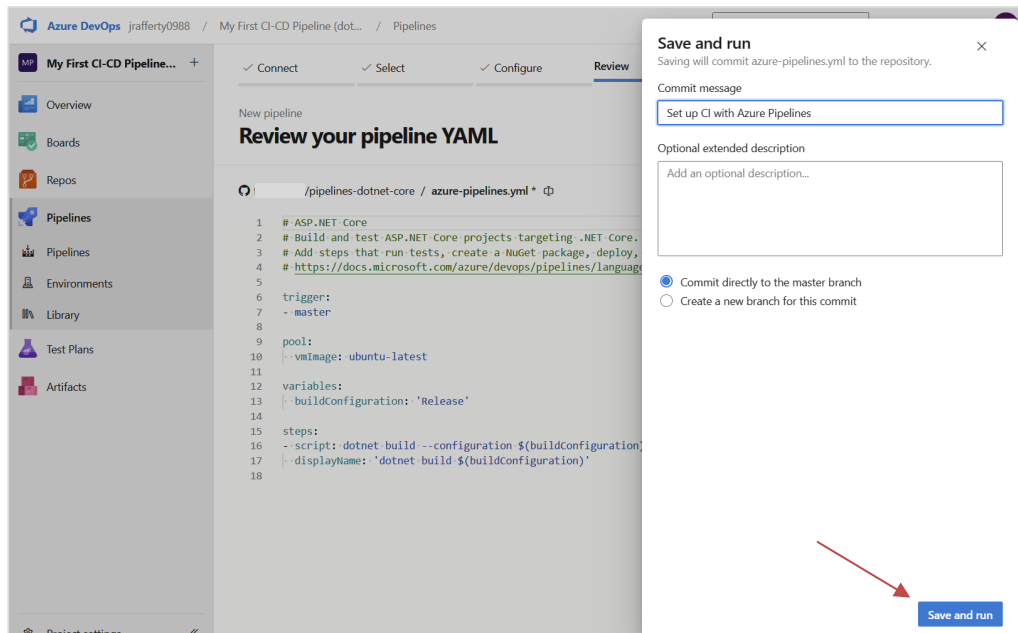


A pipeline templated will be presented, examine its contents and try and comprehend what it is specifying. Once you have some understanding, click Save and Run.

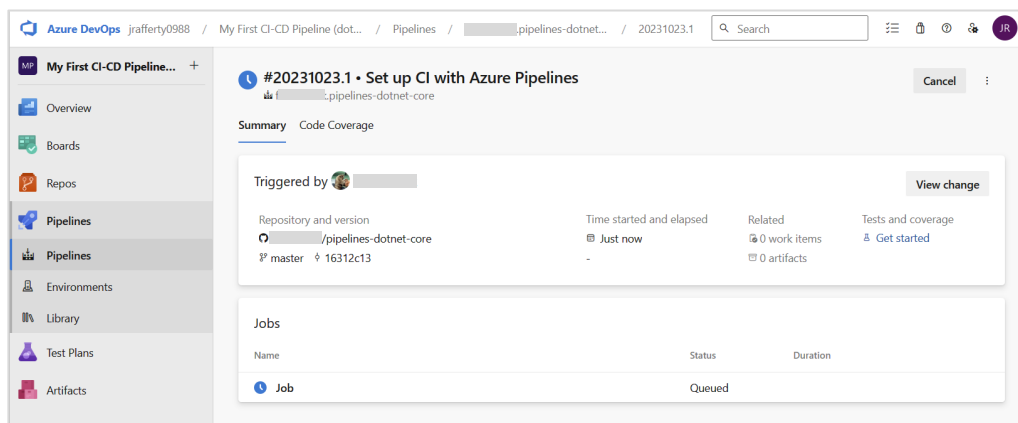


An interface will be provided requesting the commit message, the default should be fine as is the

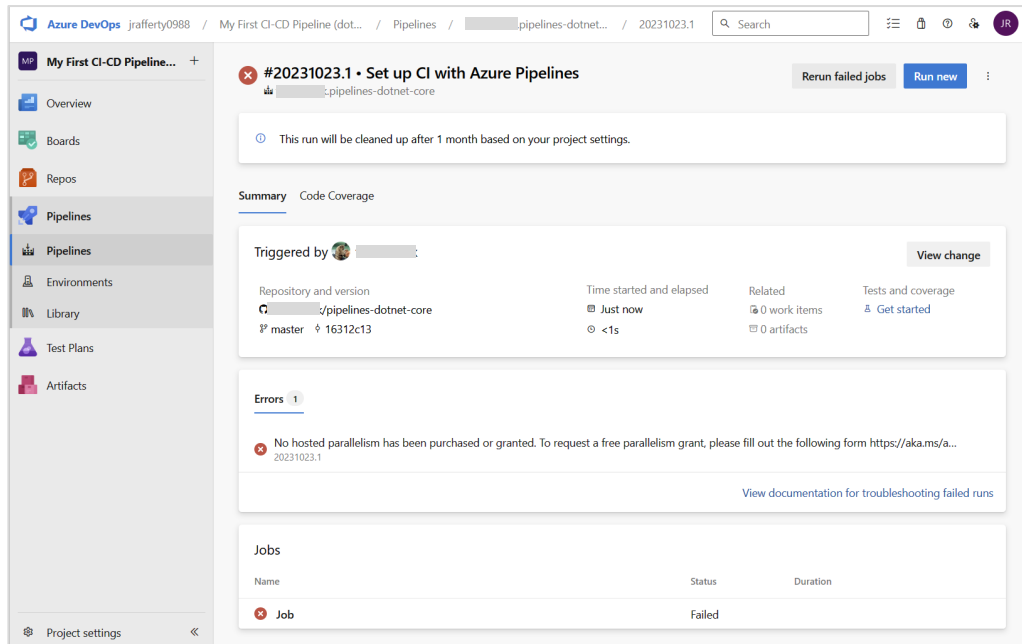
default of committing directly to the master branch. Then click save and run.



The build process will be initiated, congratulations you have established the foundations of a CI/CD pipeline.



Note: if you have not been provided a parallelism grant yet then it will fail until this has been provided, the outcome of this circumstance is presented below. Once you have the grant approved, then the build process will succeed.



Azure DevOps jrafferty0988 / My First CI-CD Pipeline (dot... / Pipelines / pipelines-dotnet... / 20231023.1

#20231023.1 • Set up CI with Azure Pipelines Rerun failed jobs Run new

This run will be cleaned up after 1 month based on your project settings.

Summary Code Coverage

Triggered by [User]

Repository and version: pipelines-dotnet-core master 16312c13

Time started and elapsed: Just now <1s

Related: 0 work items 0 artifacts

Tests and coverage: Get started

Errors 1

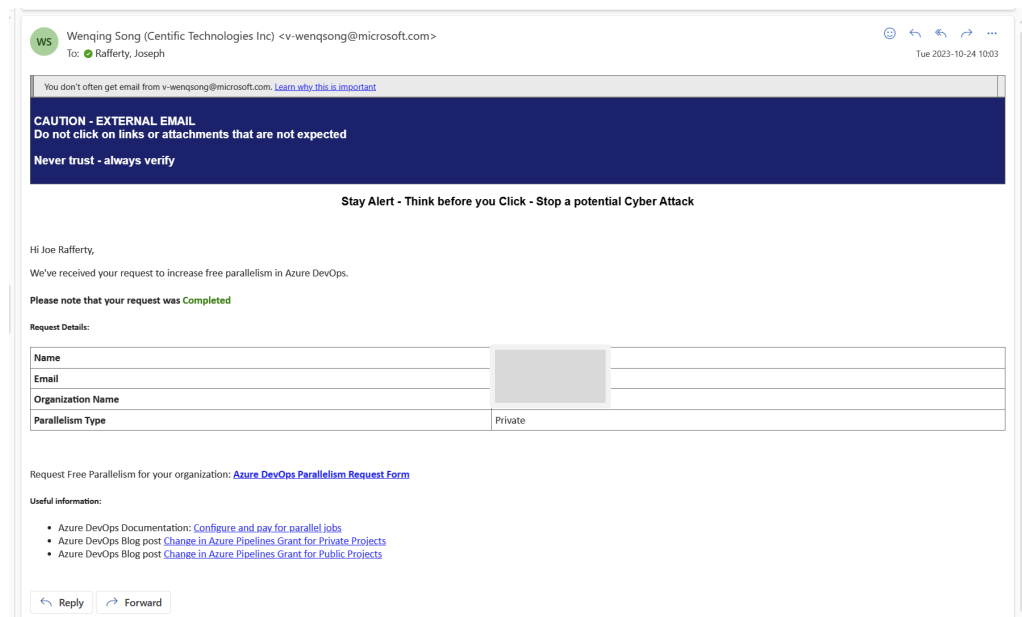
No hosted parallelism has been purchased or granted. To request a free parallelism grant, please fill out the following form <https://aka.ms/a...>

View documentation for troubleshooting failed runs

Jobs

Name	Status	Duration
Job	Failed	

When your Parallelism grant has been approved you will receive an email, similar to that shown below.



Wenqing Song (Centific Technologies Inc) <v-wenqsong@microsoft.com>
To: Rafferty, Joseph

Tue 2023-10-24 10:03

You don't often get email from v-wenqsong@microsoft.com. [Learn why this is important](#)

CAUTION - EXTERNAL EMAIL
Do not click on links or attachments that are not expected
Never trust - always verify

Stay Alert - Think before you Click - Stop a potential Cyber Attack

Hi Joe Rafferty,

We've received your request to increase free parallelism in Azure DevOps.

Please note that your request was **Completed**

Request Details:

Name	
Email	
Organization Name	
Parallelism Type	Private

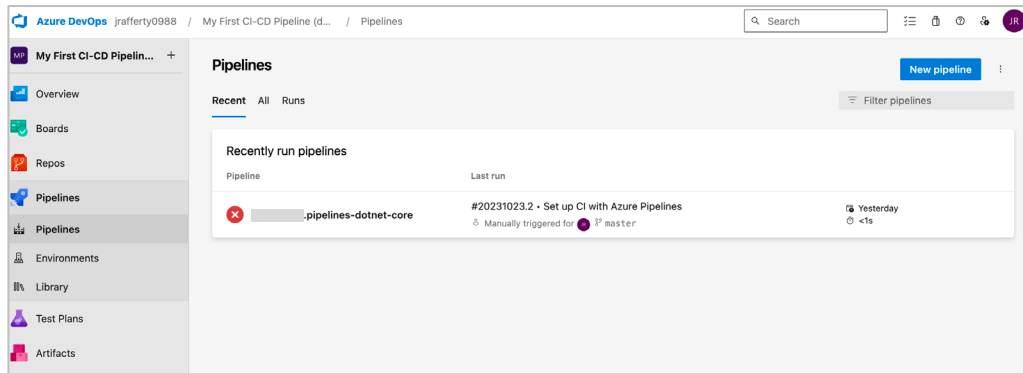
Request Free Parallelism for your organization: [Azure DevOps Parallelism Request Form](#)

Useful information:

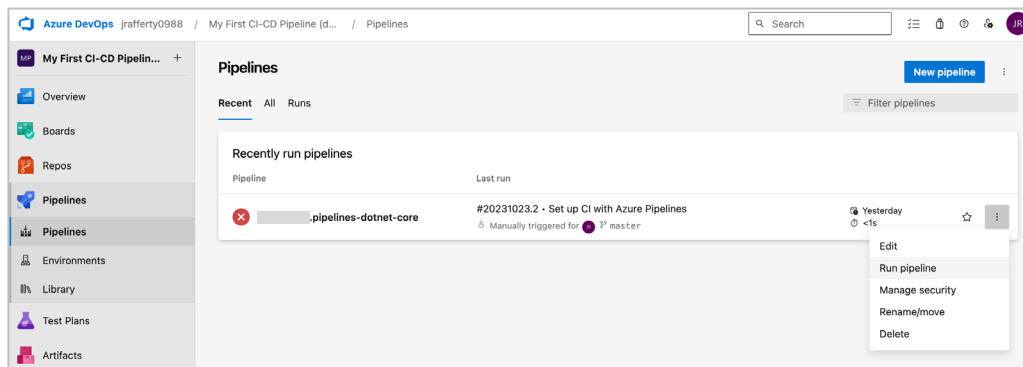
- Azure DevOps Documentation: [Configure and pay for parallel jobs](#)
- Azure DevOps Blog post [Change in Azure Pipelines Grant for Private Projects](#)
- Azure DevOps Blog post [Change in Azure Pipelines Grant for Public Projects](#)

Reply Forward

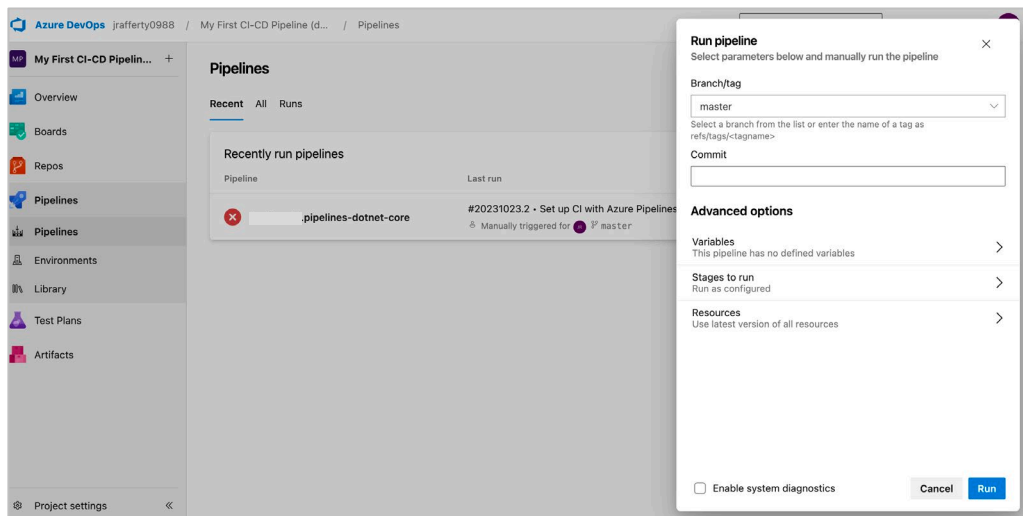
After the grant has been approved, view the DevOps organisation and then open the relevant project. Then navigate to the previously stalled pipeline. Such as that shown below.



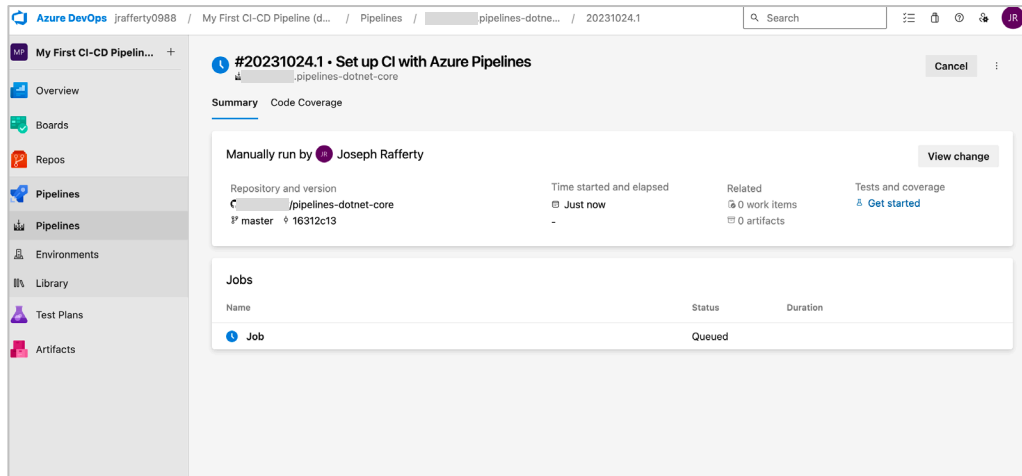
Open the options by interacting with the  button and then selecting run pipeline.



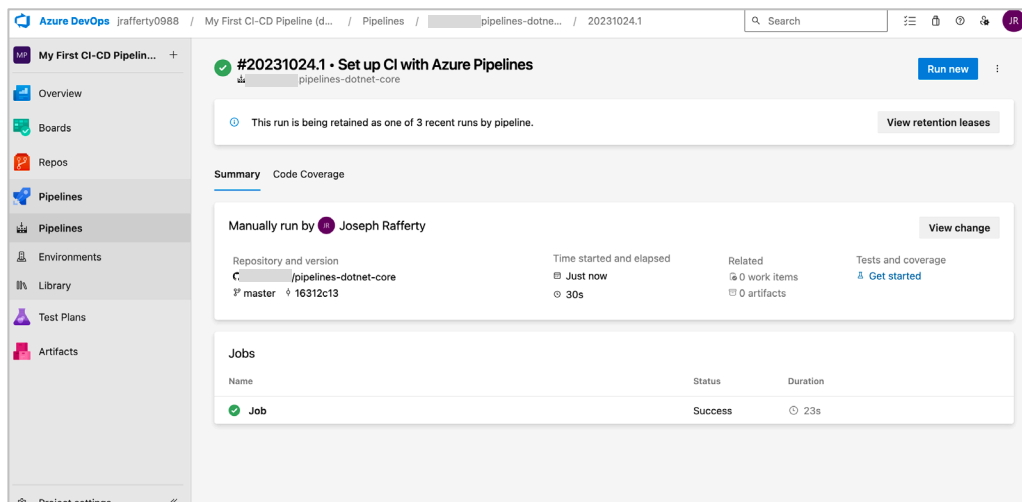
When presented with the dialog, click the run button.



The Job will be initiated and will run as appropriate.

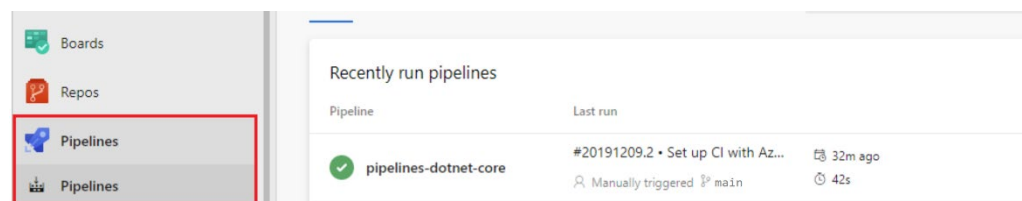


After some time, the Job will be completed.



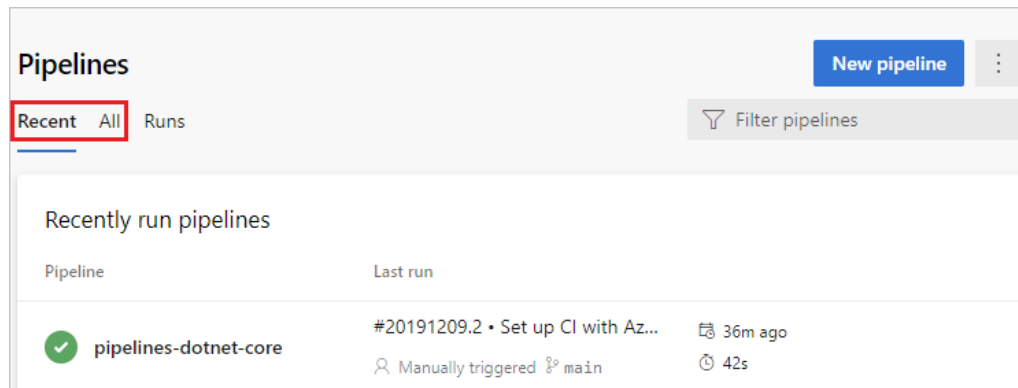
View and manage your pipelines

You can view and manage your pipelines by choosing **Pipelines** from the left-hand menu to go to the pipelines landing page.

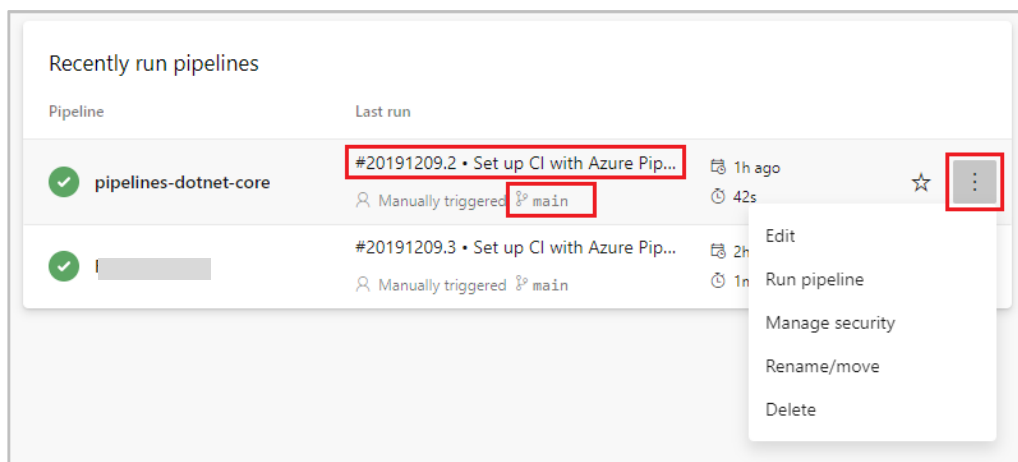


From the pipelines landing page you can view pipelines and pipeline runs, create and import pipelines, manage security, and drill down into pipeline and run details.

Choose **Recent** to view recently run pipelines (the default view), or choose **All** to view all pipelines.



Select a pipeline to manage that pipeline and [view the runs](#). Select the build number for the last run to view the results of that build, select the branch name to view the branch for that run, or select the context menu to run the pipeline and perform other management actions.



Select **Runs** to view all pipeline runs. You can optionally filter the displayed runs.

Pipelines New pipeline ⋮

Recent All **Runs** ⌵

Filter by keywords State Repository Requested for Tags X

All pipeline runs

Description	Stages	
Set up CI with Azure Pipelines #20191209.2 on pipelines-dotnet-core main d4964...	✓	44m ago 42s
Set up CI with Azure Pipelines #20191209.1 on pipelines-dotnet-core main d4964...	✓	53m ago 46s
Set up CI with Azure Pipelines #20191209.3 on FabrikamFiber main 2b4b23c	✓-✓	1h ago 1m 13s
Set up CI with Azure Pipelines #20191209.2 on FabrikamFiber main 2b4b23c	✓-✓	1h ago 1m 55s
Set up CI with Azure Pipelines #20191209.1 on FabrikamFiber main 2b4b23c	✓-✓	1h ago 1m 6s

Select a pipeline run to view information about that run.

You can choose to **Retain** or **Delete** a run from the context menu. For more information on run retention, see [Build and release retention policies](#).

Stages

✓	51m ago 42s	⋮
✓		
✓-✓	2h ago 1m 13s	

Retain
Delete

View pipeline details

The details page for a pipeline allows you to view and manage that pipeline.

Runs

Branches

Analytics

Description	Stages	
#20191209.3 Set up CI with Azure Pipelines Manually triggered main 2b4b23c	✓-✓	🕒 54m ago 🕒 1m 13s
#20191209.2 Set up CI with Azure Pipelines Manually triggered main 2b4b23c	✓-✓	🕒 55m ago 🕒 1m 55s
#20191209.1 Set up CI with Azure Pipelines Individual CI main 2b4b23c	✓-✓	🕒 56m ago 🕒 1m 6s

Choose **Edit** to edit your pipeline. For more information, see [YAML pipeline editor](#). You can also edit your pipeline by modifying the **azure-pipelines.yml** file directly in the repository that hosts the pipeline.

View pipeline run details

From the pipeline run summary you can view the status of your run, both while it is running and when it is complete.


#20191210.2 Update azure-pipelines.yml for Azure Pipe...
Run new
⋮

on FabrikamFiber

Summary
Environments

Triggered by 

 FabrikamFiber  main b2f795e

 Today at 12:56 PM

 Duration: 1m 9s

 Tests: [Get started](#)

 Changes: 2 commits

 Work items: 1 linked

 Artifacts: 1 published

Stages
Jobs


Build

 1 job completed 41s


Deploy

 1 job completed 13s

 1 artifact

From the summary pane you can view job and stage details, download artifacts, and navigate to linked commits, test results, and work items.

Jobs and stages

The jobs pane displays an overview of the status of your stages and jobs. This pane may have multiple tabs depending on whether your pipeline has stages and jobs, or just jobs. In this example, the pipeline has two stages named **Build** and **Deploy**. You can drill down into the pipeline steps by choosing the job from either the **Stages** or **Jobs** pane.

Stages <u>Jobs</u>			
Name	Status	Stage	Duration
✓ Build	Success	Build	🕒 40s
✓ DeployWeb	Success	Deploy	🕒 10s

Choose a job to see the steps for that job.

← Jobs in run #20191...

FabrikamFiber

Build

- ✓ Build 40s
 - ✓ Initialize job 1s
 - ✓ Checkout 3s
 - ✓ CmdLine 2s
 - ✓ Component Detect 32s
 - ✓ Post-job: Checkout <1s
 - ✓ Finalize Job <1s
- Deploy
 - > ✓ DeployWeb 10s
- Finalize build
 - ✓ Report build status <1s

✓ Build

1 Pool: Azure Pipelines

2 Image: Ubuntu-16.04

3 Agent: Hosted Agent

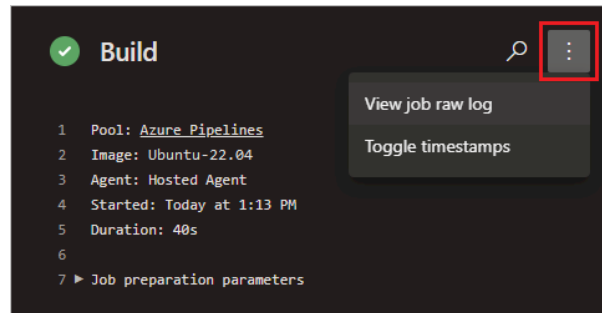
4 Started: Today at 1:13 PM

5 Duration: 40s

6

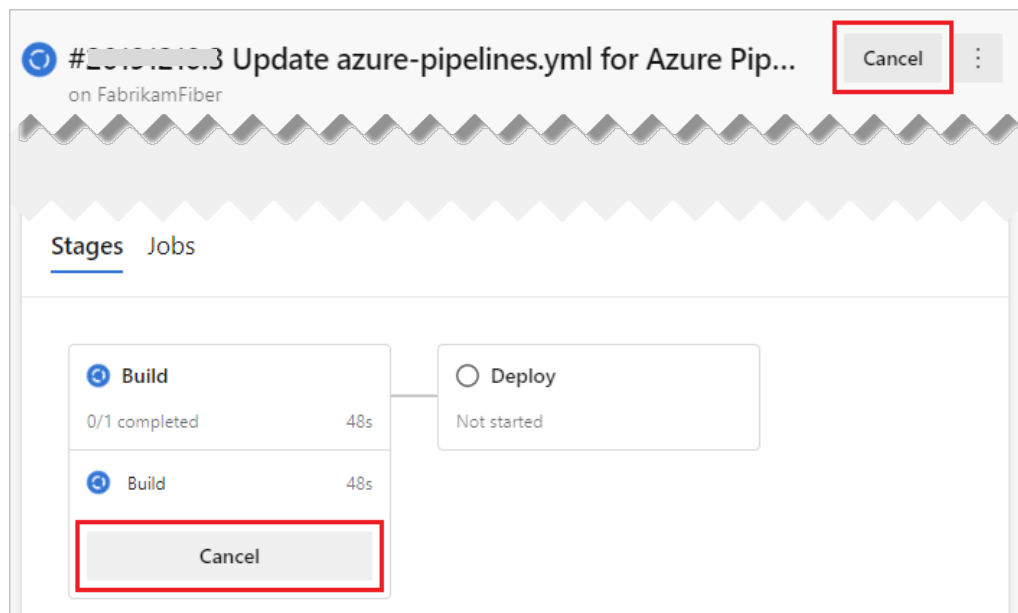
7 ▶ Job preparation parameters

From the steps view, you can review the status and details of each step. From the **More actions**  you can toggle timestamps or view a raw log of all steps in the pipeline.



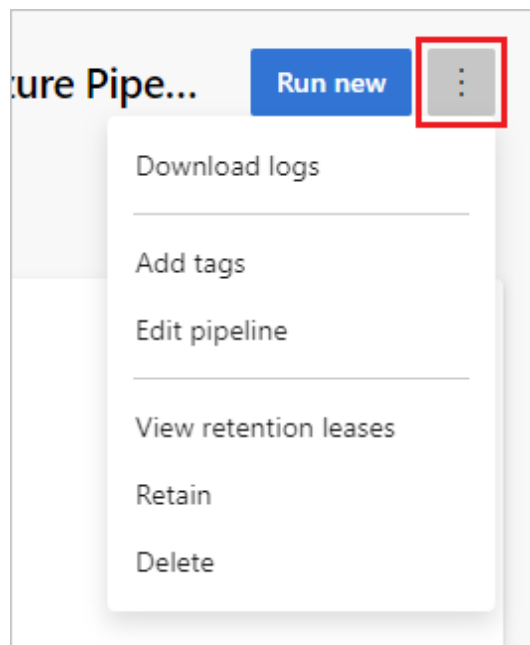
Cancel and re-run a pipeline

If the pipeline is running, you can cancel it by choosing **Cancel**. If the run has completed, you can re-run the pipeline by choosing **Run new**.



Pipeline run more actions menu

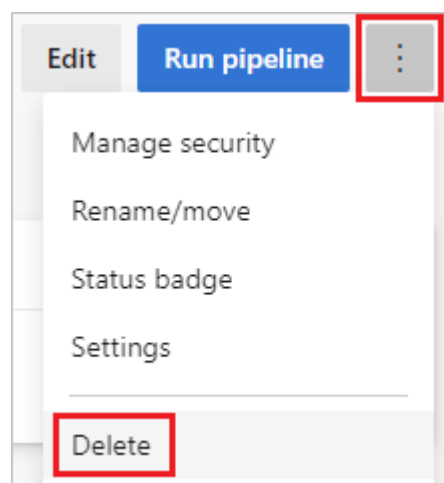
From the **More actions**  menu you can download logs, add tags, edit the pipeline, delete the run, and configure [retention](#) for the run.



NOTE: You can't delete a run if the run is retained. If you don't see **Delete**, choose **Stop retaining run**, and then delete the run. If you see both **Delete** and **View retention releases**, one or more configured retention policies still apply to your run. Choose **View retention releases**, delete the policies (only the policies for the selected run are removed), and then delete the run.

Clean up

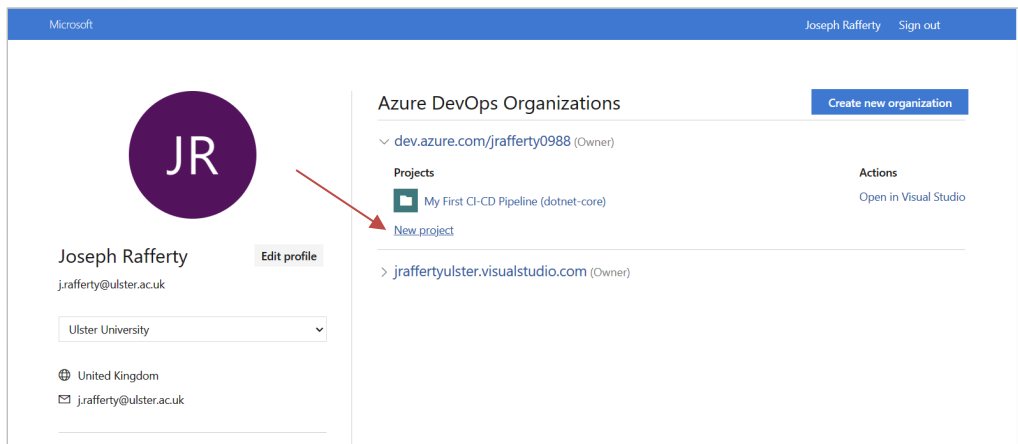
If you created any test pipelines, they are easy to delete when you are done with them. To delete a pipeline, navigate to the summary page for that pipeline, and choose **Delete** from the ... menu at the top-right of the page. Type the name of the pipeline to confirm, and choose **Delete**.



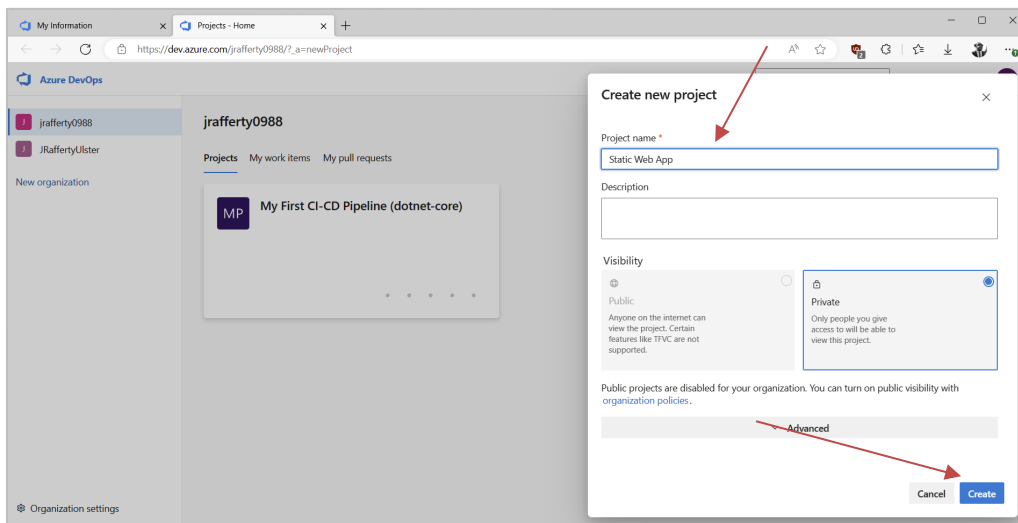
Creating a CI/CD pipeline for a new codebase

Following the establishment of an existing CI/CD pipeline, this task will show you how to create a new GitHub repository from a codebase and integrate it into a pipeline.

Firstly, we need to create a new project for a static website. Open the Azure DevOps Organizations interface located @ <https://aex.dev.azure.com/me?mkt=en-US>



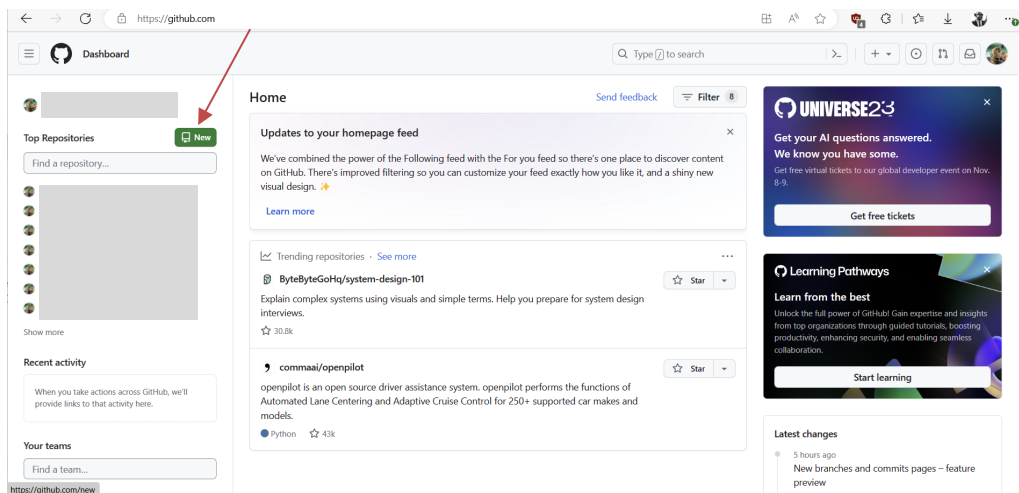
Create a new project, in this case it has been labelled **Static Web App**.



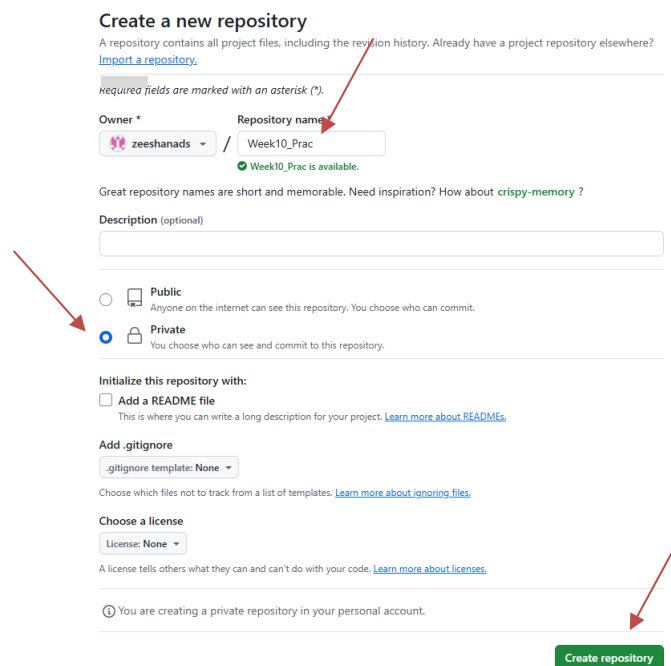
Obtain the source code from blackboard, it is under learning materials and is called [Week 9 Practical_Web App Files 24_25.zip](#).

Once downloaded, take a note of the location of the zip archive.

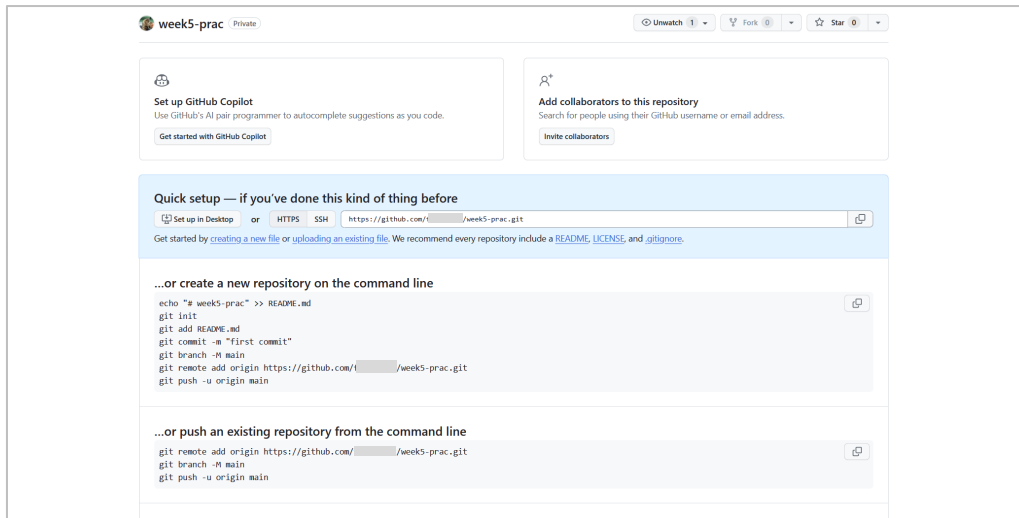
Next, use a web browser to navigate to GitHub using the account you have previously associated with Azure create a new Repository.



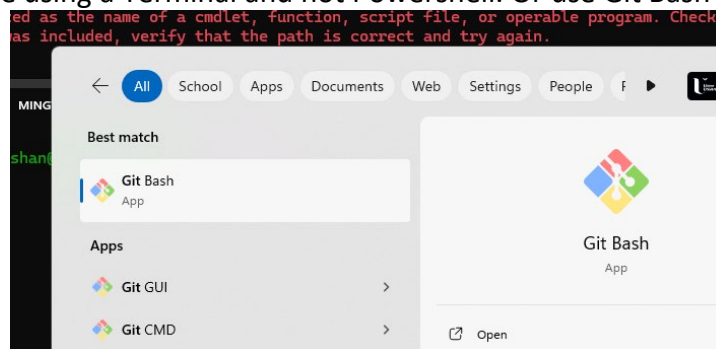
Provide a new name for the repository, set the visibility to Private, and then initiate its creation. In this case the repository name **Week10_Prac** was chosen.



Once this is completed, you will be presented with a confirmation of the creation and some key information to facilitate setup.

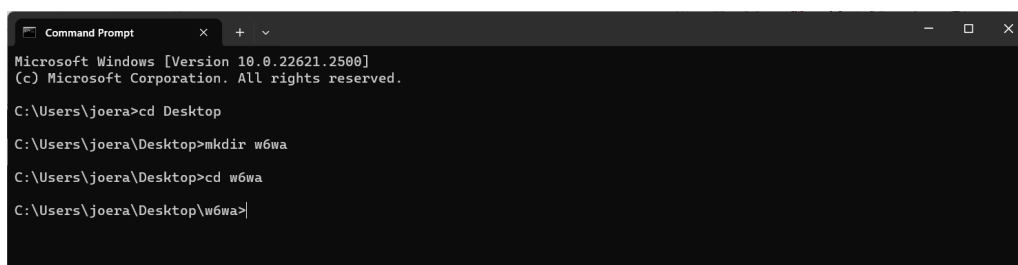


Following this stage, open a terminal so that you can use the GIT command line tools. Please ensure that you are using a Terminal and not Powershell. Or use Git Bash



Using the terminal, change your working directory to the Desktop by issuing a **cd Desktop** command. This will work if you are in your **\Users\<<USERNAME>>** working directory, which is the default. **<<USERNAME>>** is a variable and below it is the account for **joera**.

Next create a directory to hold the local copy of your repository, in this case we wish to hold this in a directory called **w10prac**). To create this directory, issue the **mkdir w10prac** command.

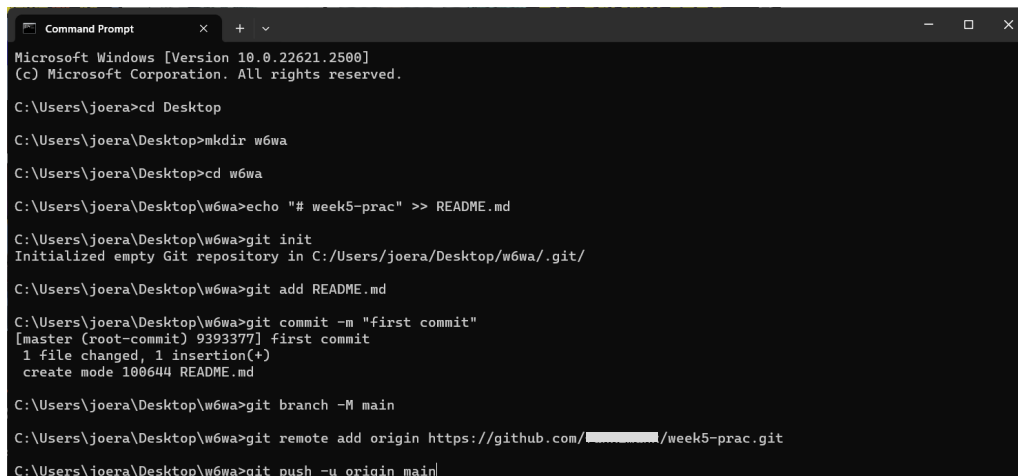


Now that a directory exists to hold the local git repository, you must initialise this folder with the necessary git data structures. To do this, issue the following series of commands. Note, the two variables **<<GITHUB_ACCOUNT_PATH>>** and **<<GITHUB_REPOSITORY_NAME>>** which you must change to reflect your account path and your repository

name.

The account path is generally a GitHub user name and the repository name is as you specified when creating the repository, in this practical it was **week10_prac** (make sure you have Git installed)

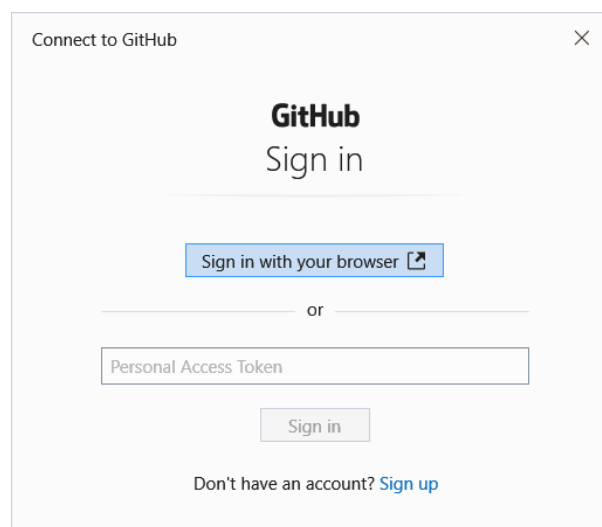
```
echo "# week10_prac" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/<<GITHUB_ACCOUNT_NAME>>/<<GITHUB_REPOSITORY_NAME>>.git
git push -u origin master
```



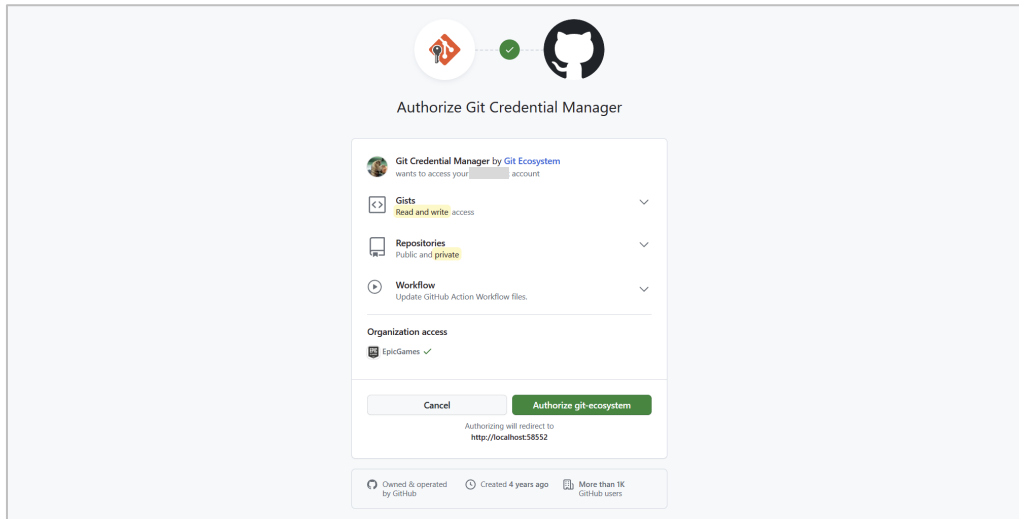
```
Microsoft Windows [Version 10.0.22621.2500]
(c) Microsoft Corporation. All rights reserved.

C:\Users\joera>cd Desktop
C:\Users\joera\Desktop>mkdir w6wa
C:\Users\joera\Desktop>cd w6wa
C:\Users\joera\Desktop\w6wa>echo "# week5-prac" >> README.md
C:\Users\joera\Desktop\w6wa>git init
Initialized empty Git repository in C:/Users/joera/Desktop/w6wa/.git/
C:\Users\joera\Desktop\w6wa>git add README.md
C:\Users\joera\Desktop\w6wa>git commit -m "first commit"
[master (root-commit) 9393377] first commit
1 file changed, 1 insertion(+)
create mode 100644 README.md
C:\Users\joera\Desktop\w6wa>git branch -M main
C:\Users\joera\Desktop\w6wa>git remote add origin https://github.com/[redacted]/week5-prac.git
C:\Users\joera\Desktop\w6wa>git push -u origin main
```

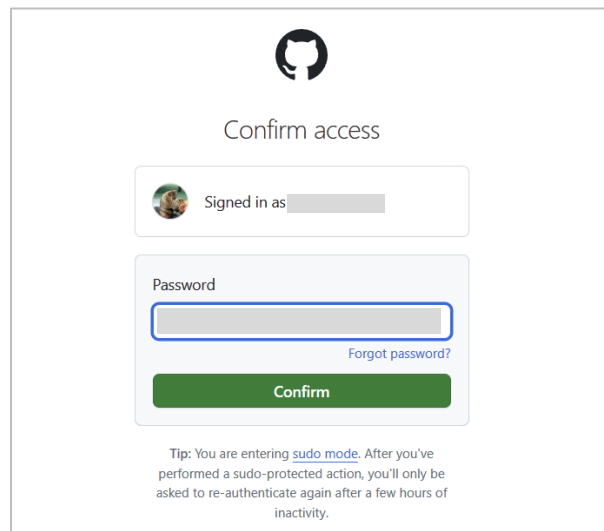
During this process you may be prompted to sign into your GitHub account. Select Sign in with your browser or use a Personal Access Token if you have one. In this practical we selected to sign in with the browser.



When signing in with the browser, an authorisation prompt will be presented, as shown below. Authorise the access as required by this practical.



You may need to subsequently confirm access using your password.



Following this authorisation, the empty git repository will be pushed to the remote GitHub repository.

```
C:\Users\joera\Desktop\w6wa>git remote add origin https://github.com/[redacted]/week5-prac.git

C:\Users\joera\Desktop\w6wa>git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 226 bytes | 226.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/[redacted]/week5-prac.git
 * [new branch]      main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.

C:\Users\joera\Desktop\w6wa>
```

Leave this terminal open as you will need it later

Following this initialisation and pushing of the repository to the remote server, locate the [Week 9 Practical_Web App Files 24_25.zip](#) folder which you previously downloaded.

Extract the contents to the [w10prac](#) folder you have created on the desktop. Please note that the target w10prac directory is no longer empty as it has been initialised with git version control file structures.

Return to the open terminal and issue the following command: **git add .**
This command will stage all files in the repository, including the newly extracted ones.

```
C:\Users\joera\Desktop\w6wa>git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 226 bytes | 226.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/[redacted]/week5-prac.git
 * [new branch]    main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.

C:\Users\joera\Desktop\w6wa>git add .\
```

Next issue the following command to commit the code to the repository.

git commit -m "initial commit"

In this case the message/comment is set to “initial commit”.

```
Command Prompt
C:\Users\joera\Desktop\w6wa>git commit -m "initial commit"
[main 23bc9cd] initial commit
10 files changed, 16510 insertions(+)
create mode 100644 W6WebApp\W6WebApp\CSS\bootstrap.css
create mode 100644 W6WebApp\W6WebApp\CSS\bootstrap.min.css
create mode 100644 W6WebApp\W6WebApp\CSS\bootstrap.min.css.map
create mode 100644 W6WebApp\W6WebApp\JS\app.js
create mode 100644 W6WebApp\W6WebApp\JS\bootstrap.bundle.min.js
create mode 100644 W6WebApp\W6WebApp\JS\bootstrap.bundle.min.js.map
create mode 100644 W6WebApp\W6WebApp\JS\bootstrap.js
create mode 100644 W6WebApp\W6WebApp\JS\bootstrap.js.map
create mode 100644 W6WebApp\W6WebApp\JS\jquery-3.6.0.min.js
create mode 100644 W6WebApp\W6WebApp\index.html

C:\Users\joera\Desktop\w6wa>
```

Next issue the following command to push the newly committed changes to the remote origin server, your repository on GitHub.

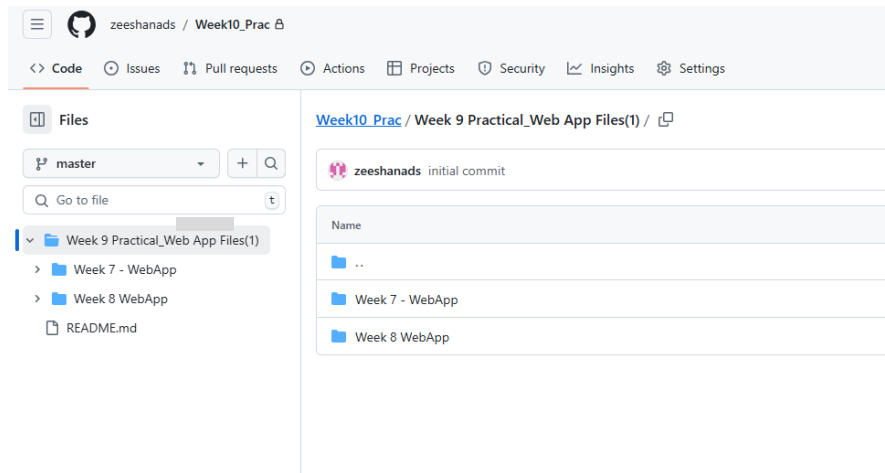
git push origin master

```
C:\Users\joera\Desktop\w6wa>git push origin main
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 24 threads
Compressing objects: 100% (15/15), done.
Writing objects: 100% (16/16), 354.59 KiB | 7.24 MiB/s, done.
Total 16 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/[redacted]/week5-prac.git
 9393377..23bc9cd main -> main

C:\Users\joera\Desktop\w6wa>
```

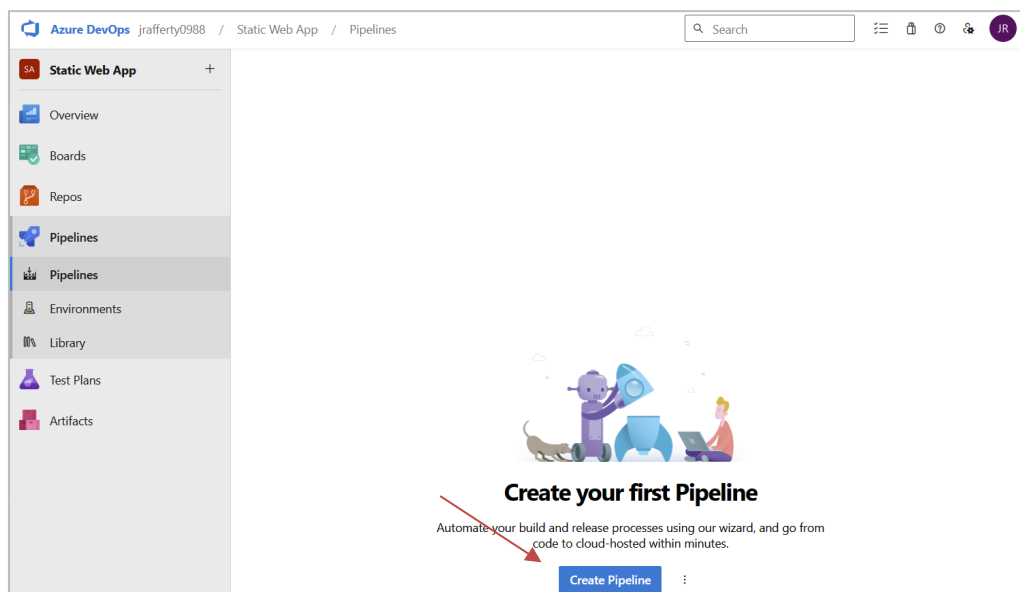
Open GitHub in the web browser and Navigate to the repository, You should notice that the files are now present.

Also note that the code committed is in a nested subdirectory. This is bad practice and should be avoided.

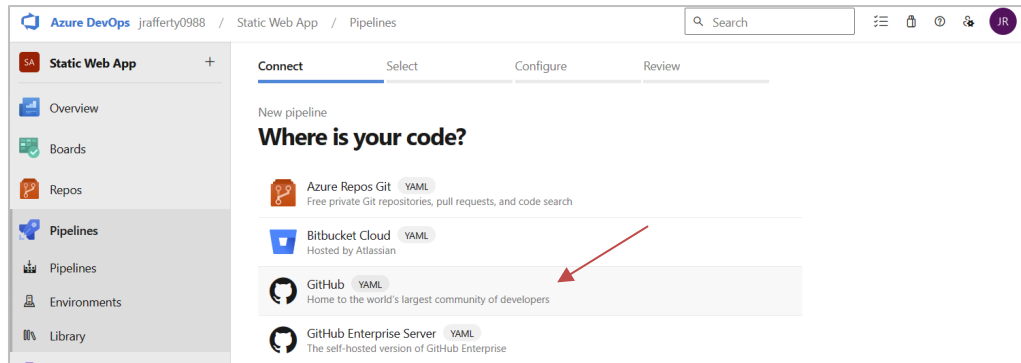


Return to the Azure DevOps Organizations view at <https://aex.dev.azure.com/me?mkt=en-US> and open the new project, previously called Static Web App.

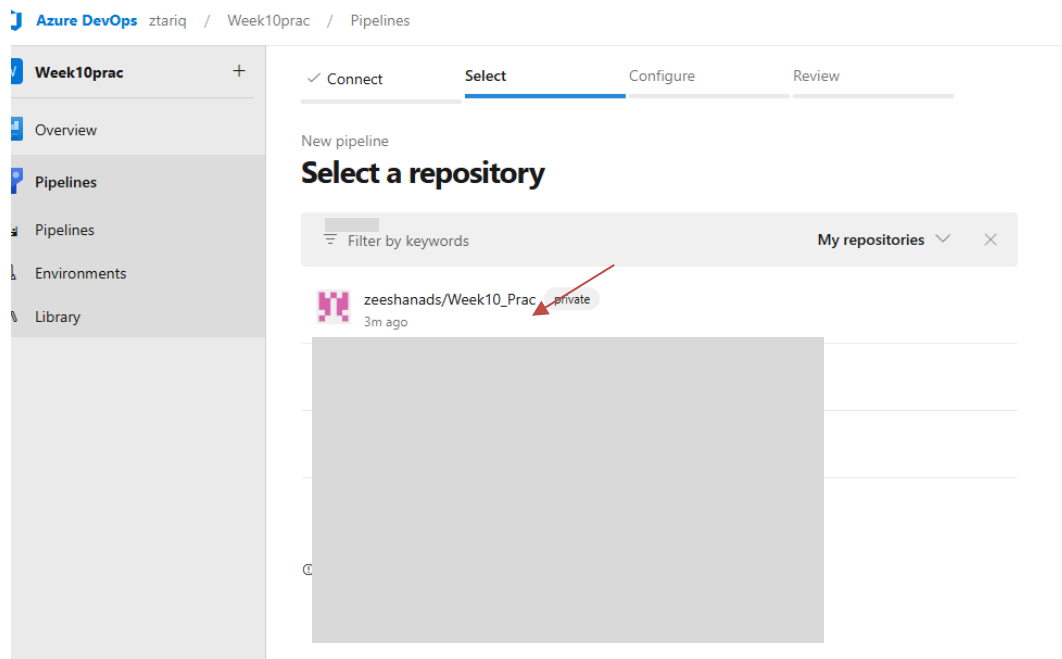
Go to the pipelines interface and create a pipeline.



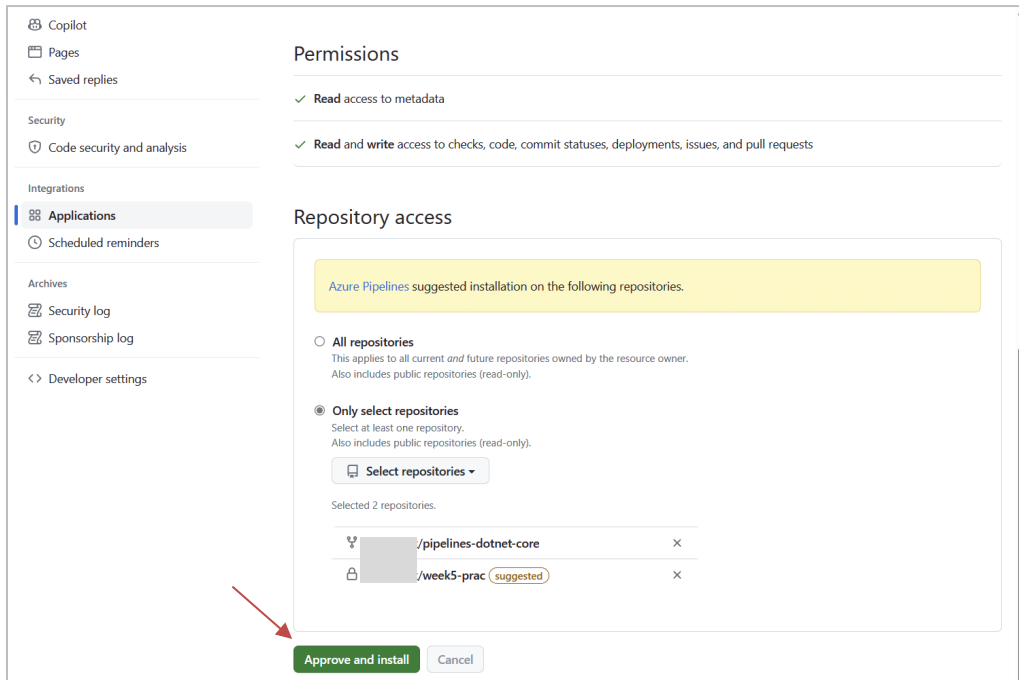
Connect a GitHub repository to the pipeline.



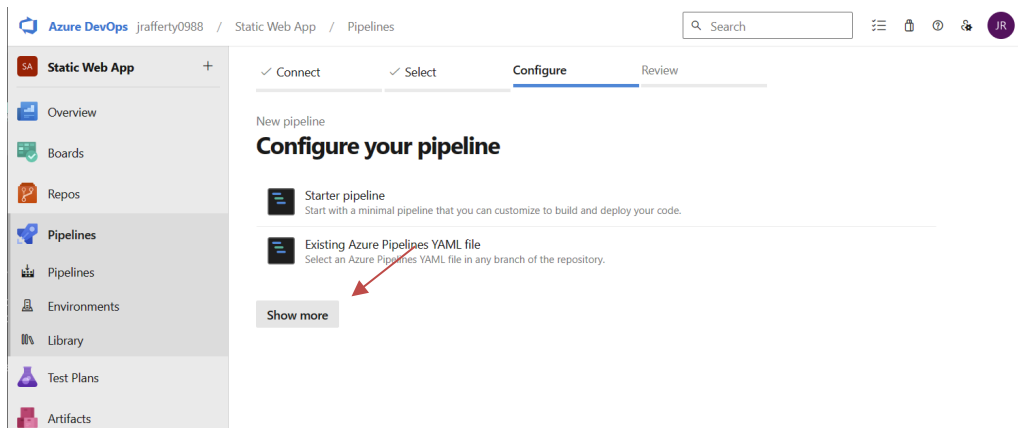
Select the repository you have created, in this practical it was called week5-prac



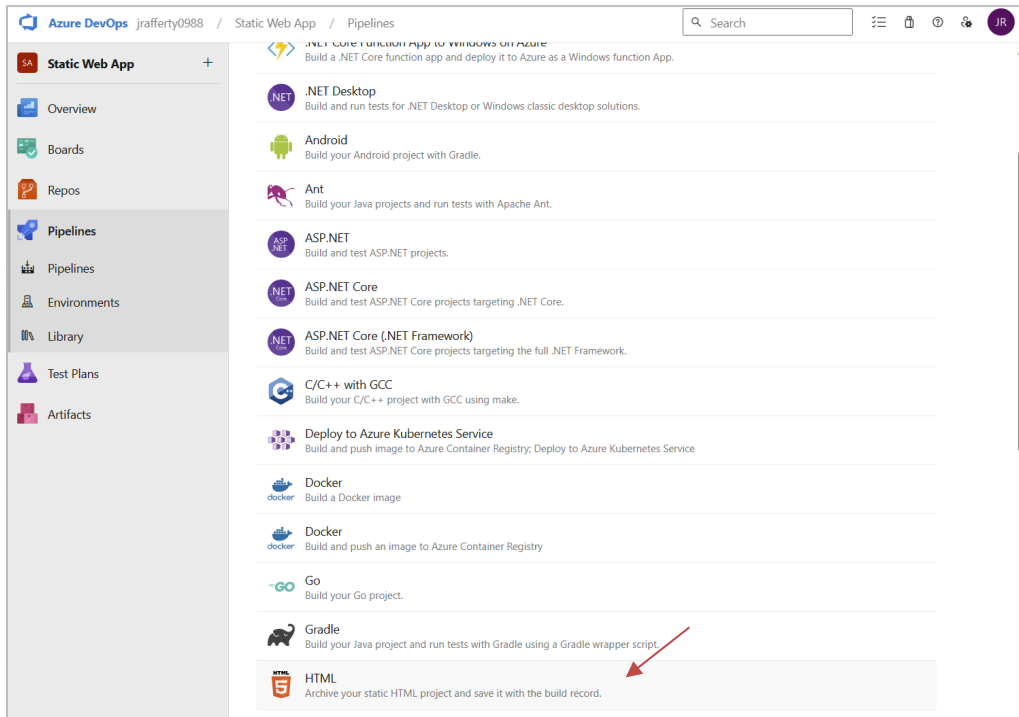
Provide the necessary permissions.



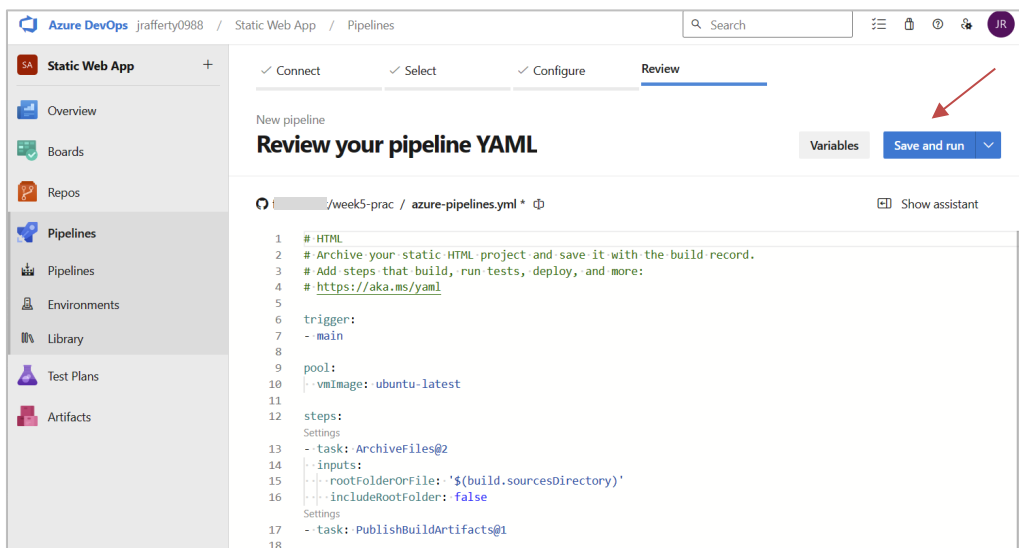
Next, continue to configure the pipeline, the option we want is not presented so select show more.



As this is a static HTML project, navigate to that option then select it.



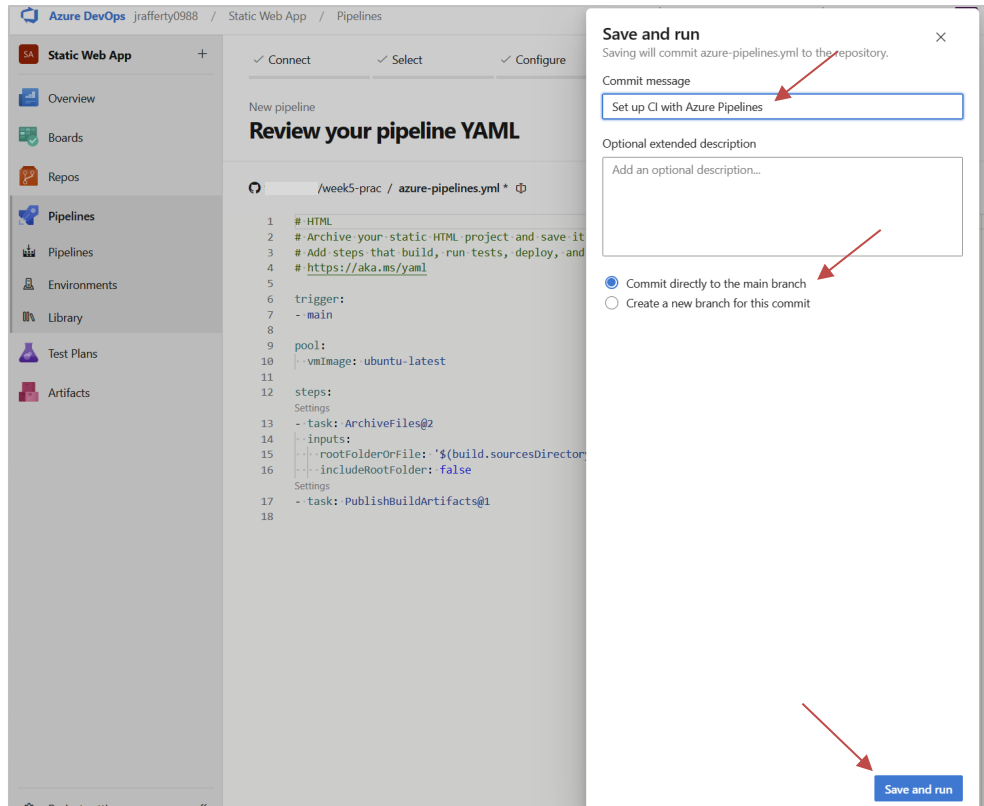
A template will be presented, for the purposes of this practical don't deviate from the default. Click Save and Run



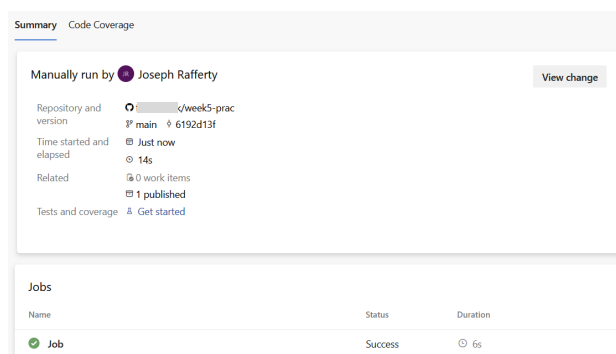
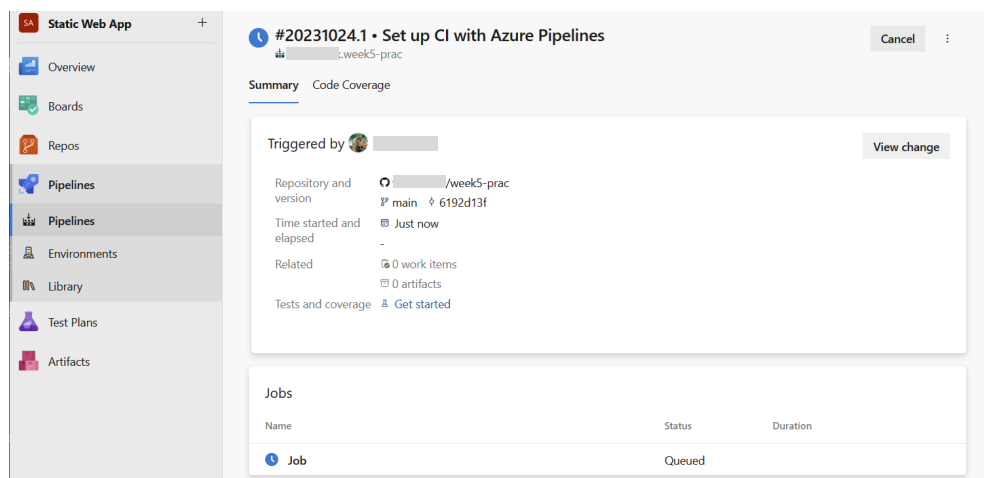
This YAML file will be created and committed to the Git repository. You will therefore be asked if you would like this stored in the main branch or in a new one.

For the purpose of this practical, we will store this in the main branch.

Provide a related commit name, specify to store this in the main branch and then click initiate the save and run process.



Note that the pipeline will run and build.



Further Learning

In this practical you have established CI/CD pipelines from git repositories in a basic manner.

You are encouraged to read further into the process to understand all the facets of this operation.

To do this, the most appropriate source is <https://learn.microsoft.com/en-us/azure/devops/pipelines/?view=azure-devops>

It is particularly encouraged to read this <https://learn.microsoft.com/en-us/azure/devops/pipelines/ecosystems/dotnet-core?view=azure-devops&tabs=dotnetfive> and <https://learn.microsoft.com/en-us/azure/app-service/deploy-azure-pipelines?view=azure-devops&tabs=yaml> and then expand the first section of this practical to **automatically deploy to a web app**.

Clean up resources

You can delete Azure App Service and related resources when you don't need them anymore. Use the **Delete** functionality on the DevOps Starter dashboard.